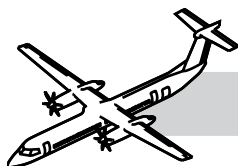


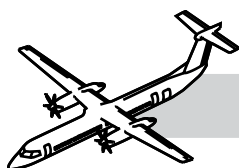
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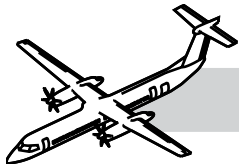
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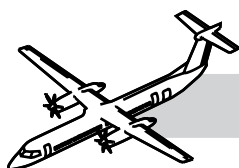
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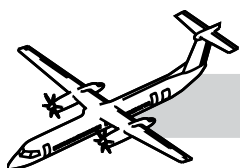
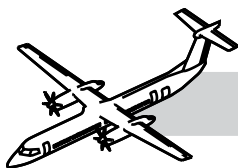


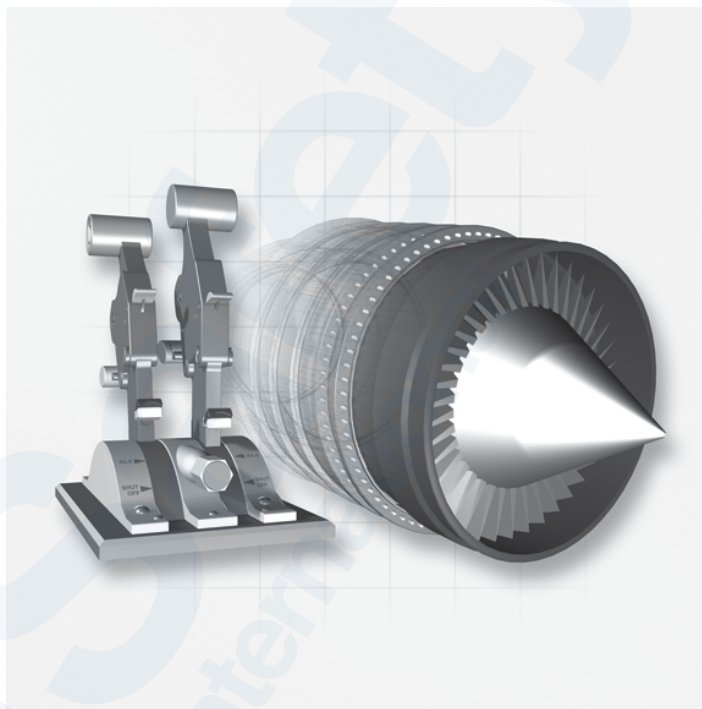
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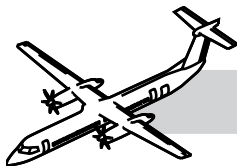
79-00-00 INTRODUCTION

The engine oil system supplies filtered oil to the engine for lubrication cooling and control.

GENERAL

The system is comprised of the sub-systems that follow:

- Oil Distribution
- Oil Pressure and Scavenge
- Oil Indicating.



79-20-00 OIL DISTRIBUTION SYSTEM

GENERAL

The purpose of the oil distribution system is to supply, filtered oil to the engine for lubrication of moving parts.

SYSTEM DESCRIPTION

The system is a wet-sump system. Oil is cooled by an aircraft mounted air-cooled oil cooler.

The system supplies a constant flow of clean filtered oil to the turbo-machinery and reduction gearbox bearings to cool and lubricate the running shafts, gears, and component surfaces.

The oil is contained in a tank that is an integral part of the LP compressor case. The tank has a sight glass, for viewing of the oil quantity, and a filler neck and cap for replenishing the oil supply.

A second, smaller, tank is in the reduction gearbox. This tank is supplied with oil from the main tank. The oil pumps are assembled together as one unit, and the entire assembly is installed in a bore on the LP compressor case.

The system has these three sub-systems:

- Pressure system that supplies oil to the reduction gearbox and the turbo machinery
- Scavenge system that returns the used oil to the tank
- Vent and breather system that vents the bearing cavities and removes any air trapped in the scavenged oil.

Filters in the pressure and scavenge systems remove contaminants from the oil. The filter housing contains impending bypass switches, which sense the pressure differential.

When a filter is becoming blocked, the switches send a signal to the related FADEC and the EMS to warn of an impending bypass.

79-22-00 OIL PRESSURE AND SCAVENGE SYSTEM

GENERAL

The Oil pressure system supplies pressurized, filtered oil to the following components:

- Engine
- Propeller control system
- AC generator.

The scavenge system returns the oil from these components, filtering out debris as the oil is returned.

DESCRIPTION

Leading Particulars - Oil System

Refer to Figure 79-1. Oil Pressure System.

Maximum Oil Consumption:

- 0.127 US qt/Flight Hour or 0.24 lb./FH or 0.11 Kg/FH
- 1 US gal = 7.74 lb. (ref. only).

Limits:

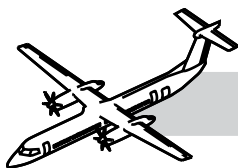
- Starting (minimum)..... -40°C
- Take-off..... 65 to 107°C
- Transient (maximum)..... 20 min.

NOTE

Starting (minimum) as per AFM is -40°C.

Oil Tank:

- Capacity..... 5.9 US gal. (22.2 litre)
- Min to max on sight glass 1.6 US qts. (1.5 litre).

**Pressure Pumps:**

- The ejector pump supplies positive pressure to the vane type pump
- It is activated by PRV bypass pressure
- The engine driven, vane type pump has positive displacement protected by a strainer.

NOTE**Cold Start Valve (CSV):**

- Limits pressure surge during cold starting
- Opens at 300 to 600 psid.

Pressure oil filter housing:

- Uses a 12 micron filter element
- The impending bypass indicator will trigger at 17 - 24 psid (signal sent to FADEC and EMU, fault code to CDS)
- The filter bypass valve opens at 34 psid (234 kPad).

No.1 bearing cavity:

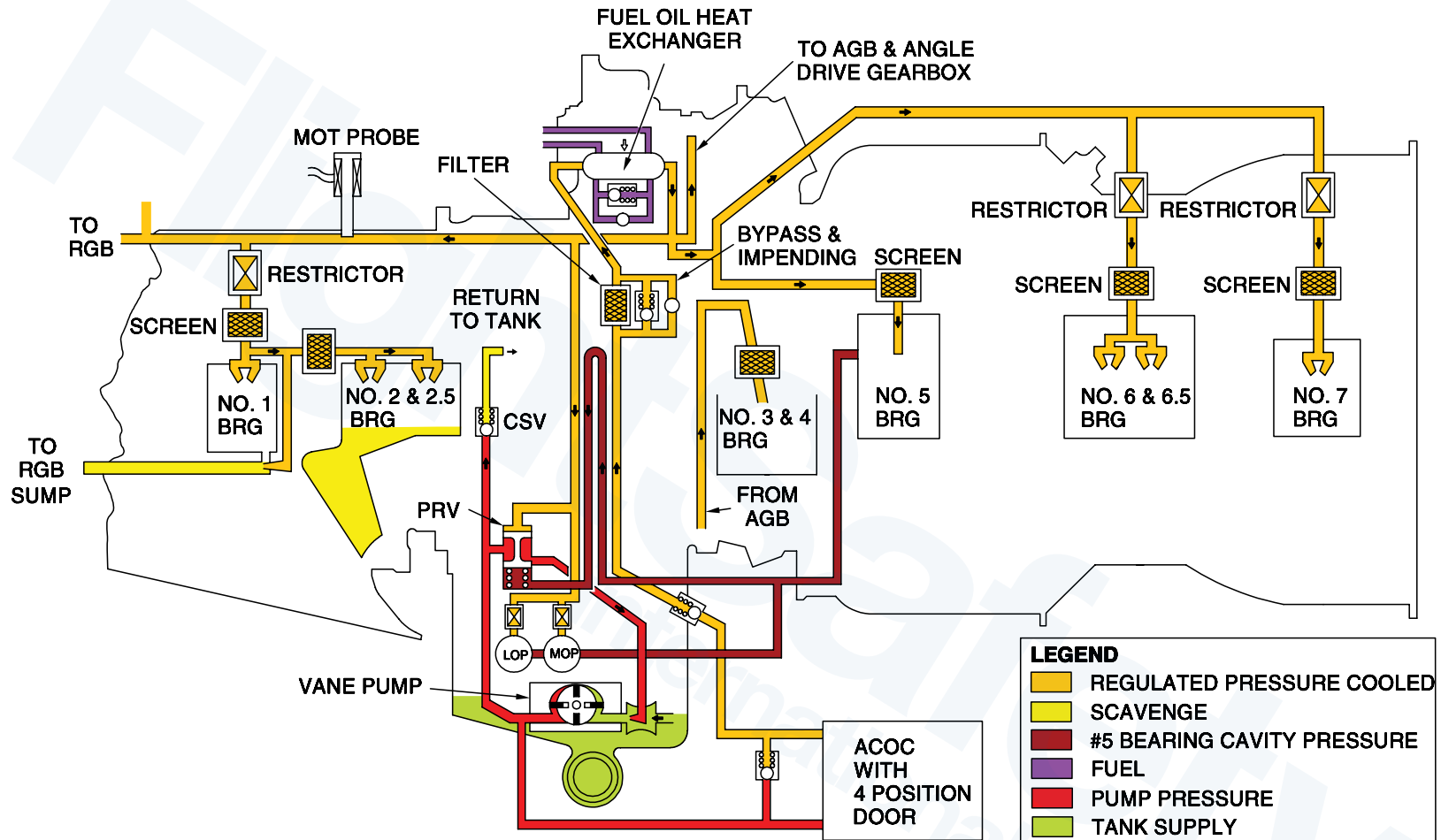
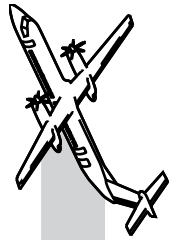
- Has two nozzles, one on each face of the bearing
- Has a restrictor to drop the pressure to compensate for low cavity air pressure.

No.2 and No.2.5 bearing cavity:

- Is an oil manifold with multiple nozzles for bearing No.2 and No.2.5
- It uses a restrictor to drop the pressure to compensate for lower cavity air pressure
- Has a last chance strainer.

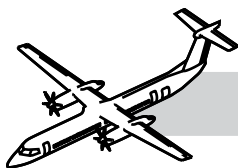
No.3 and No.4 bearing cavity:

- Is an oil manifold with multiple nozzles for bearing No.3 and No.4
- Has a last chance strainer.



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Figure 79-1. Oil Pressure System



No.5 bearing cavity:

- Is an oil manifold with multiple nozzles for bearing No.5
- Has a last chance strainer.

NOTE

No.6 and No.6.5 bearing cavity:

- Is an oil manifold with multiple nozzles for bearing No.6 and No.6.5
- It uses a restrictor to drop the pressure to compensate for lower cavity air pressure
- Has a last chance strainer.

No.7 bearing cavity:

- Is an oil manifold with multiple nozzles for bearing No.7
- It uses a restrictor to drop the pressure to compensate for lower cavity air pressure
- Has a last chance strainer.

Pressure Regulating Valve (PRV):

- Will maintain oil pressure at 61-72 psid
- The oil pressure is adjusted as a function of bearing cavity (No.5) pressure in order to provide constant oil flow at all powers
- The minimum pressure is 44 psid (303 kPad)
- It is field adjustable.

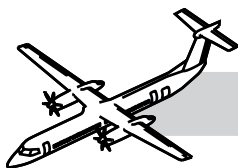
Check Valve (CV):

- It is installed at ACOC return line
- It will prevent oil from running back into the engine (usable oil reduced).



FOR TRAINING PURPOSES ONLY

Figure 79-2. Oil Scavenge System



Leading Particulars - Oil Scavenge System

Refer to:

- Figure 79-1. Oil Pressure System.
- Figure 79-3. Oil Scavenge Blowdown Valves - Removal/Installation.
- Figure 79-4. LP Compressor Case Oil Jet Pump and Strainers - Removal/Installation.

No.1 bearing cavity:

- Is scavenged by a jet pump and directs oil to RGB sump
- The jet pump activated by oil pump pressure.

No.2 and No.2.5 bearing cavity:

- Is scavenged by a vane type pump. It returns oil to tank through the deaerator.

No.3 and No.4 bearing cavity:

- Is scavenged by a vane type pump. It returns oil to tank through the deaerator.
- Has a relief valve to assist the pump at high power (blowdown) settings.

No.5 bearing cavity:

- Is scavenged by a vane type pump and returns oil to tank through the deaerator at low power settings
- Has a relief valve to assist the pump at high power (blowdown) settings. It helps prevent pump cavitation and cavity flooding.

No.6 and No.6.5 bearing cavity:

- Is scavenged by a vane type pump and returns oil to tank through the deaerator at low power settings
- Has a relief valve to assist the pump at high power (blowdown) settings. It helps prevent pump cavitation and cavity flooding.

No.7 bearing cavity:

- Is scavenged by a vane type pump and returns oil to tank through the deaerator at low power settings.

AGB:

- Is scavenged by a vane type pump and returns oil to the tank through the deaerator at low power settings.

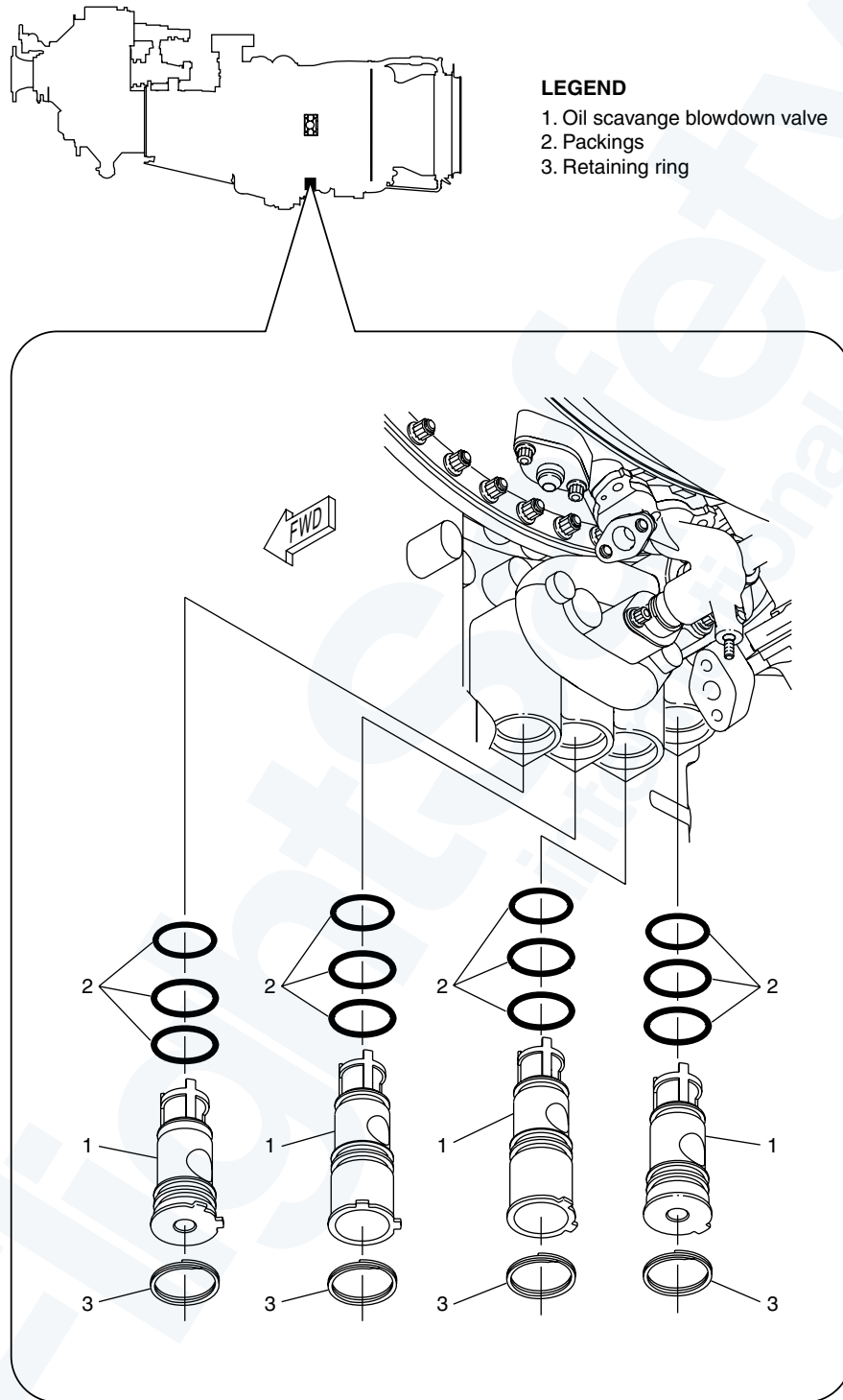
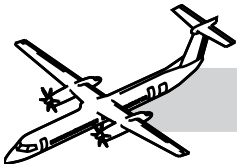


Figure 79-3. Oil Scavenge Blowdown Valves - Removal/Installation

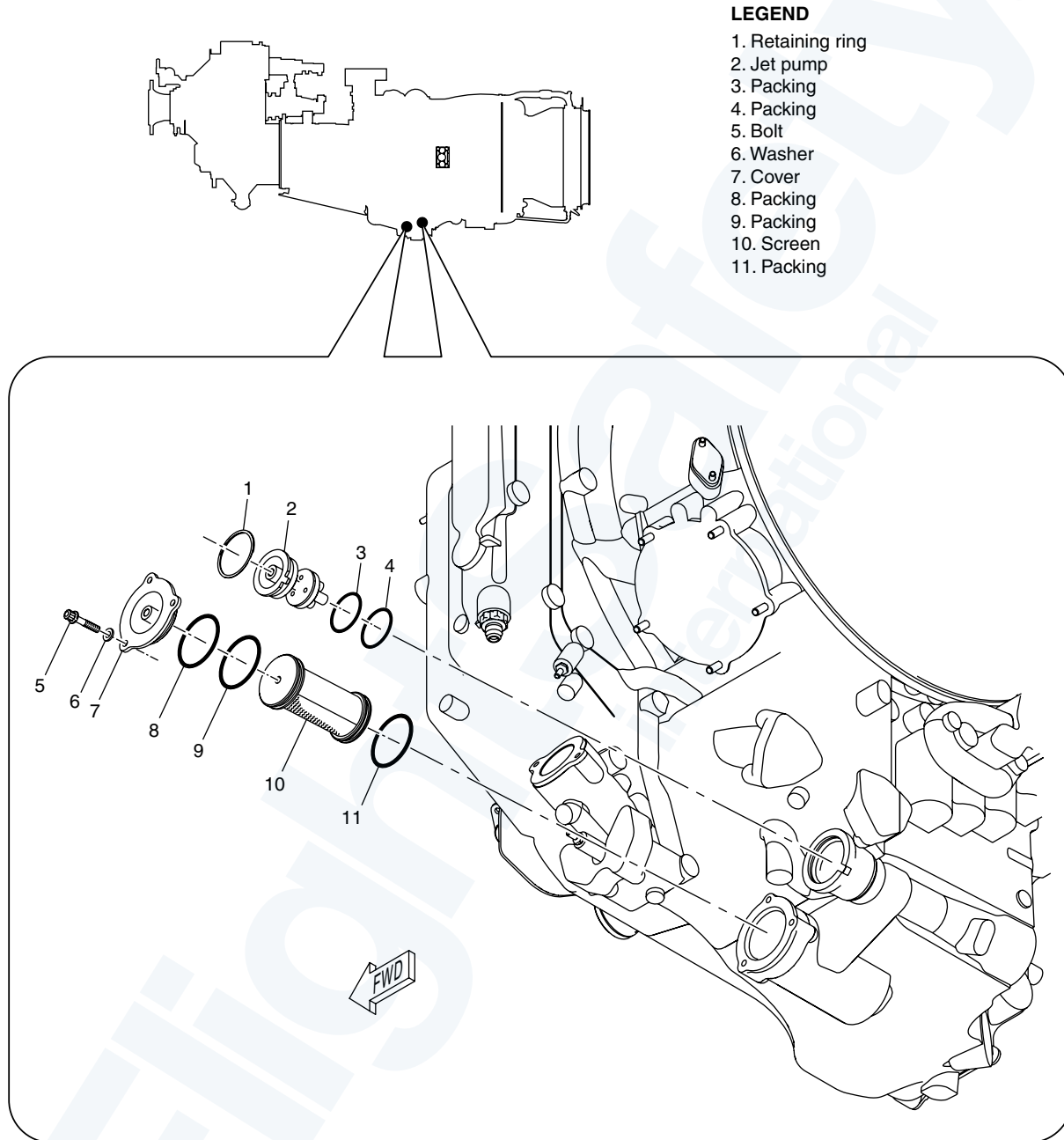
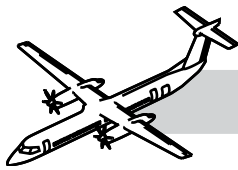


Figure 79-4. LP Compressor Case Oil Jet Pump and Strainers - Removal/Installation

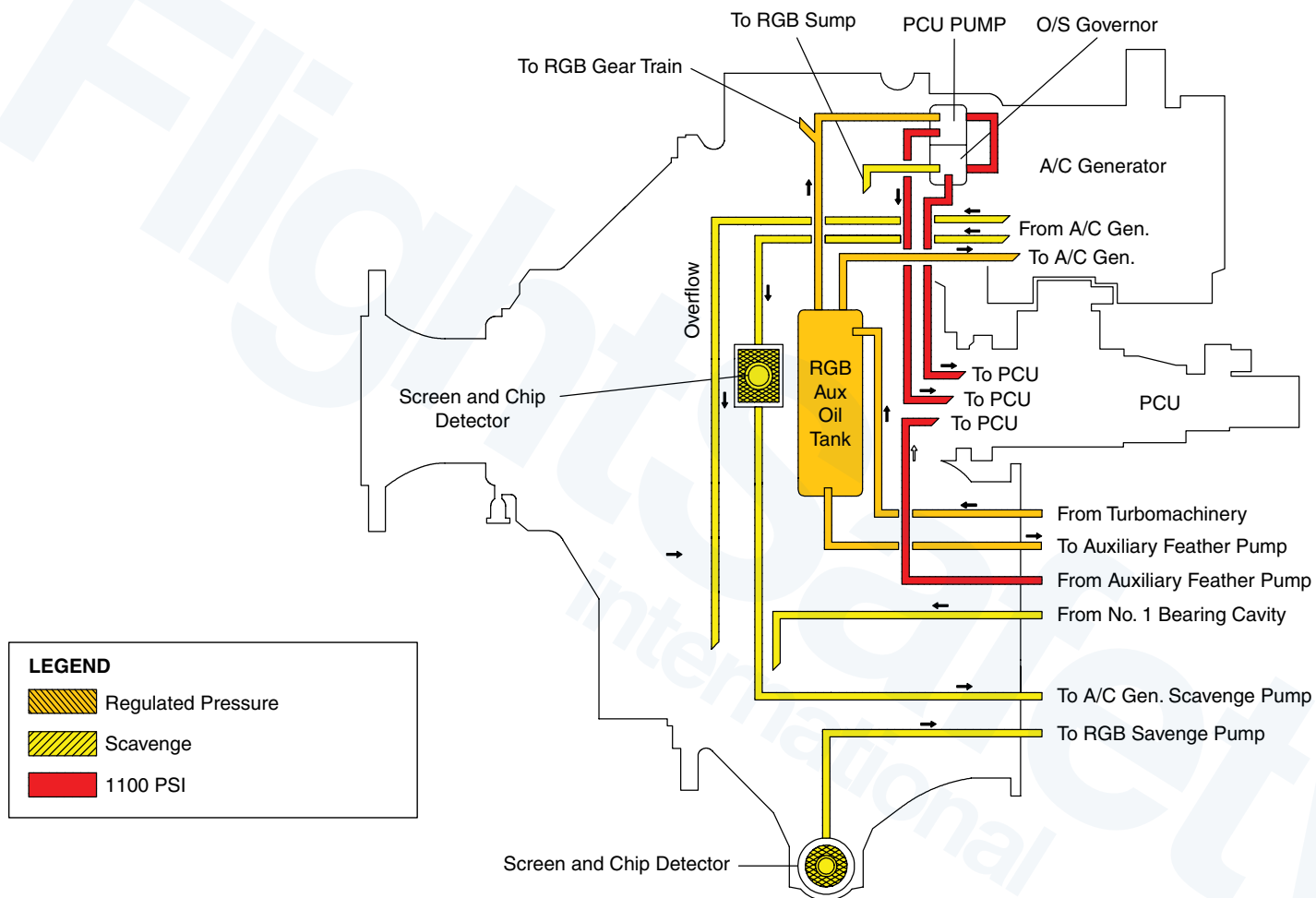
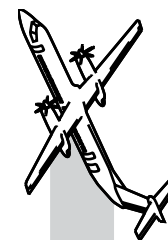
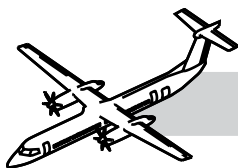


Figure 79-5. RGB Oil System



Leading Particulars - RGB Oil Pressure System

Refer to Figure 79-5. RGB Oil System.

RGB auxiliary oil tank:

- Is in the RGB rear housing
- It is a pressurized tank with capacity of 5.3 US quarts (5 liters)
- It supplies oil to:
 - RGB gear train (for lubrication)
 - AC generator (for lubrication)
 - PCU pump (to operate the propeller system)
 - Auxiliary feathering pump (to feather the propeller in an emergency during take-off)
- The Auxiliary Feathering Pump is the only LRU that can take oil from the tank in static condition.

Leading Particulars - RGB Oil Scavenge System

RGB scavenge oil filter housing:

- Has a 12 micron filter element
- Has an impending bypass indicator: at 17-24 psid (117-165 kPad)
- Has a filter bypass valve that maintains oil supply when filter becomes restricted
- It opens at 34 psid (234 kPad).

A/C generator:

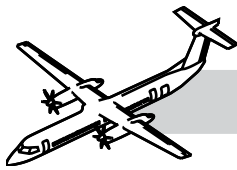
- Is scavenged by a vane type pump and returns oil to the tank through the A/C chip detector and RGB scavenge oil filter
- In case of an overflow, oil is directed to RGB sump.

Propeller system, Overspeed Governor and Gear Train:

- Oil is directed to RGB sump.

RGB sump:

- Is scavenged by a vane type pump and returns oil to the tank across the RGB chip detector and through the scavenge oil filter
- RGB scavenge oil temperature is also used to warm up the front inlet case rear lip to prevent an ice build up (anti-icing).



COMPONENT DESCRIPTION

The functions performed by the oil pressure and scavenge system are accomplished by the following components:

- PRV
- Main oil filter
- RGB scavenge oil filter
- Oil pump assembly (pressure and scavenge stages)
- Oil filler cap and tube assembly.

Main Oil Filter

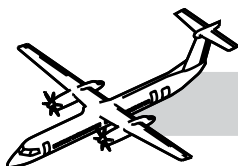
Refer to Figure 79-6. Main Oil Filter.

This filter is a 12 micron filter and is on the left side of the engine.

The filter is non-cleanable and incorporates a Dirt Alert strip that is used to analyze the debris removed from the oil.



Figure 79-6. Main Oil Filter



Pressure Regulating Valve (PRV)

Refer to Figure 79-7. Pressure Regulating Valve (PRV).

The PRV is a balanced single piston valve assembled into the left side of the low pressure compressor case.

It regulates system oil pressure from the oil pump using the No.5 bearing cavity pressure as a reference.

CAUTION

WHEN YOU MAKE AN ADJUSTMENT TO THE OIL PRESSURE, THE FINAL POSITION OF THE ADJUSTER MUST BE BETWEEN 7 AND 11 TURNS COUNTERCLOCKWISE FROM THE FULLY SEATED POSITION. IF IT IS NECESSARY TO GO OUTSIDE THESE LIMITS TO GET THE CORRECT OIL PRESSURE, THEN EITHER THE PRESSURE REGULATING VALVE OR THE PRESSURE INDICATION IS DEFECTIVE. IF YOU OPERATE THE ENGINE WITH THE OIL PRESSURE ADJUSTER OUTSIDE THE LIMITS YOU CAN CAUSE DAMAGE TO THE ENGINE.

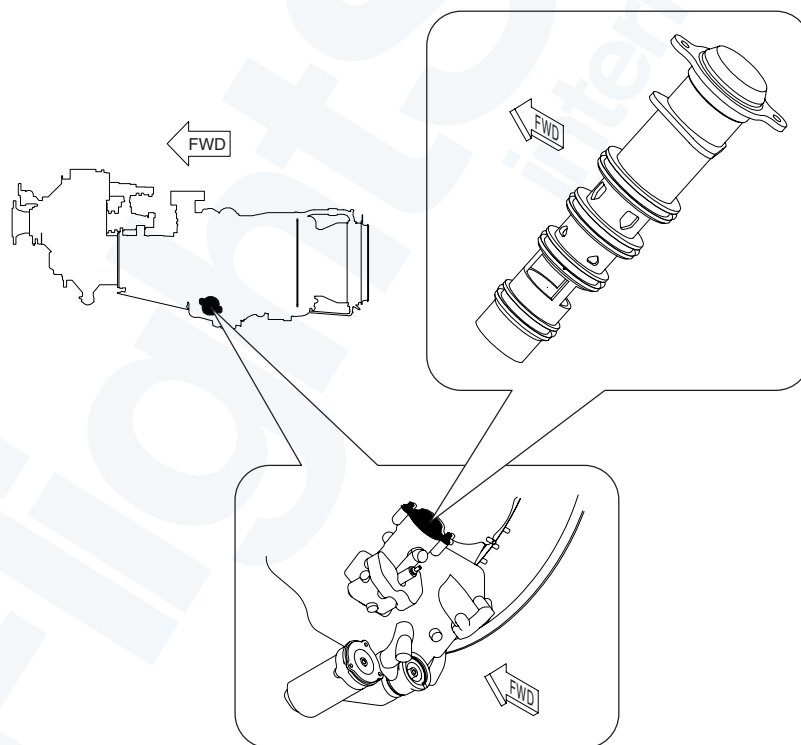
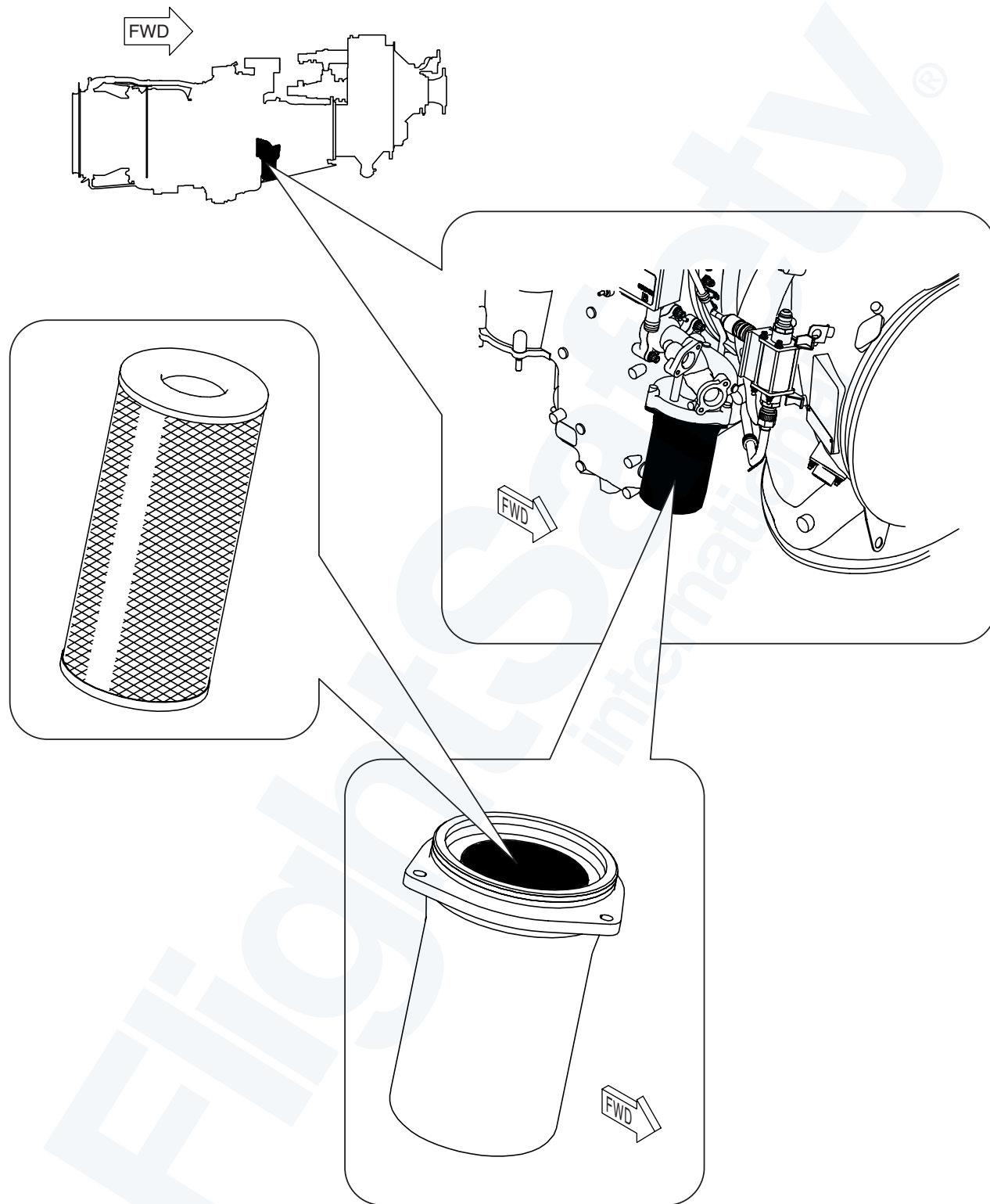
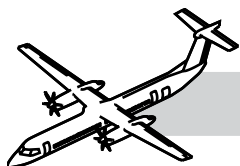
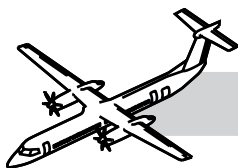


Figure 79-7. Pressure Regulating Valve (PRV)

**Figure 79-8. RGB Scavenge Oil Filter**



RGB Scavenge Oil Filter

NOTE

Refer to:

- Figure 79-8. RGB Scavenge Oil Filter.
- Figure 79-9. Dirt Alert Strip.

The RGB oil filter is on the right side of the engine. It separates any debris that may be contained in the oil that has been scavenged from the RGB.

The filter is non-cleanable and incorporates a Dirt Alert strip that is used to analyze the debris removed from the oil.

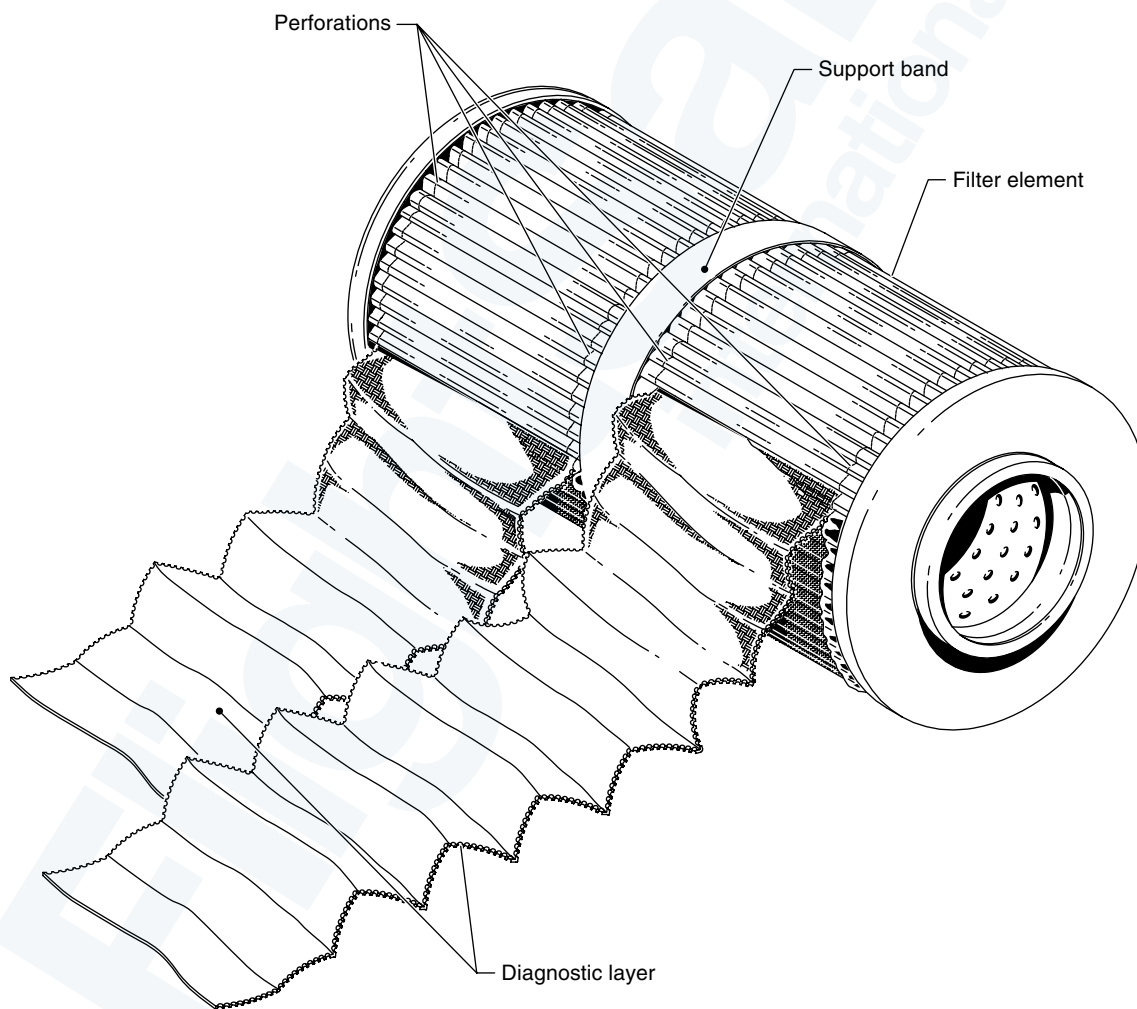
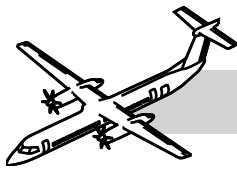


Figure 79-9. Dirt Alert Strip



Oil Pressure and Scavenge Pump Assembly

Refer to Figure 79-10. Oil Pressure and Scavenge Pump Assembly.

The oil pressure and scavenge pump assembly pressurizes and scavenges the oil in the engine oil system.

The scavenge pumps also optimize bearing cavity sealing during a sub-idle condition.

The pump assembly consists of 9 vane type pump stages and a cold start valve.

These vane pumps are contained in a pump pack and are on the right side of the engine. This pump pack is serviced as one line replaceable unit.

The cold start valve is a spring loaded poppet valve.

The cold start valve operates when very cold oil causes the oil pressure to be too high.

The valve operates at an oil pressure of 300 to 360 psi (2068 to 2482 kPad). When in operation, the valve bypasses oil back to the oil tank.

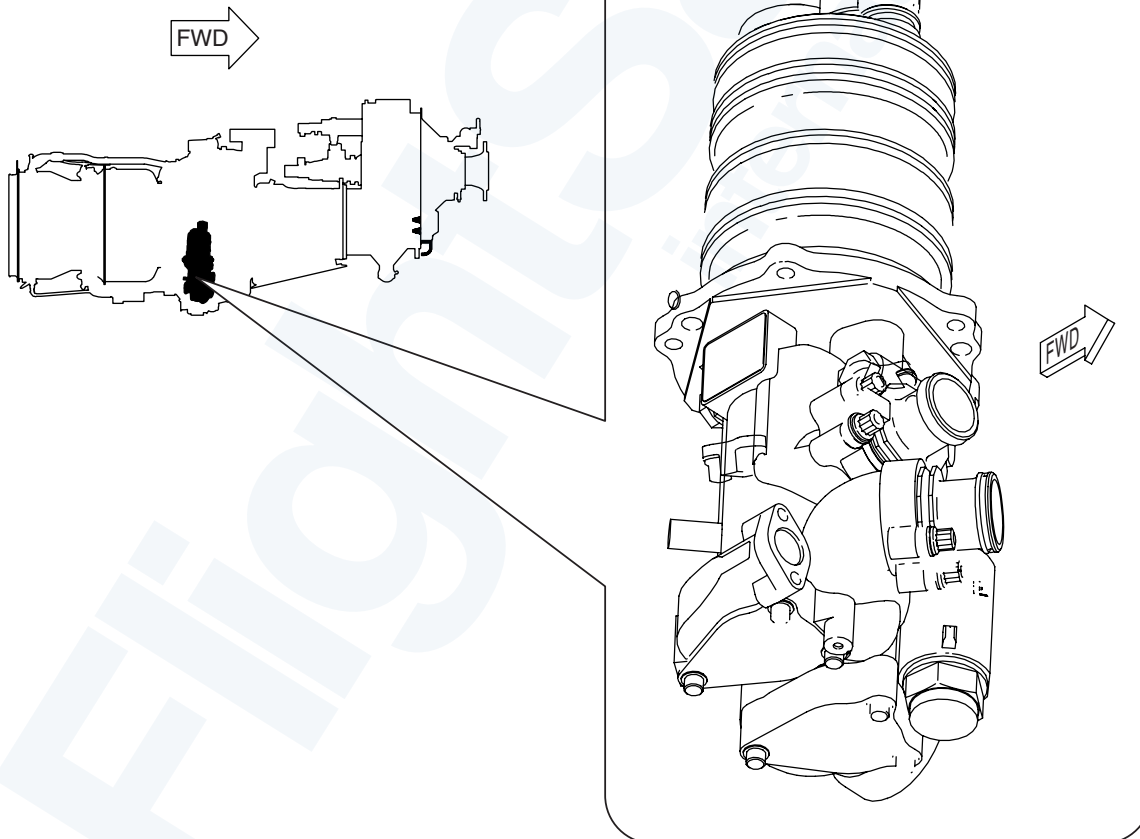
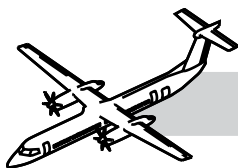


Figure 79-10. Oil Pressure and Scavenge Pump Assembly



Oil Filler Cap and Tube Assembly

Refer to Figure 79-11. Oil Filler Cap and Tube Assembly.

The oil filler cap and tube assembly is used to replenish the oil system and seal the inlet to the oil tank.

The cap is provided with a spring loaded lift flap to unlock and requires a half-turn to release it.

The oil filler has an external scupper with a drain fitting to catch any oil overflow.

Inside the filler tube is a non-return ball valve. The valve allows oil to be added and seals the filler tube to prevent oil loss in the event that the filler cap is missing.

OIL CONSUMPTION TREND MONITORING

The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

- Record the amount of oil added and the date and time
- Record the amount of hours flown between servicing
- Oil consumption calculated by the amount of oil added divided by hours flown since last servicing
- Maximum oil consumption detailed in AMM TASK 05-11-00-992-802 0.12 liters/flight hour.

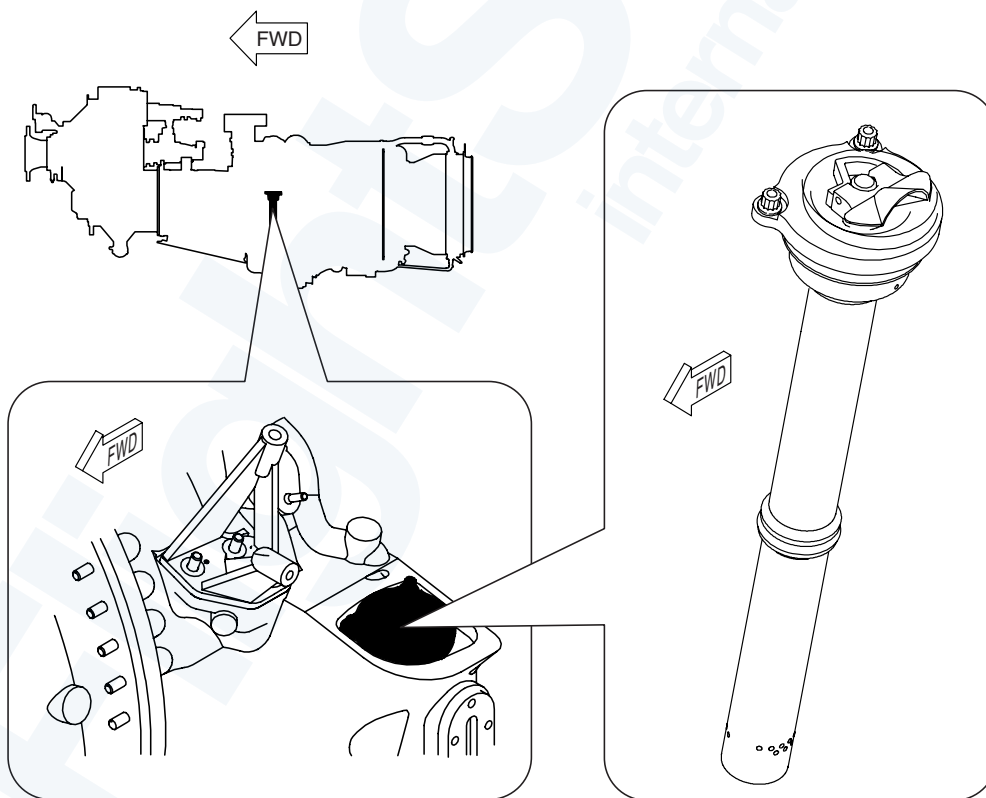
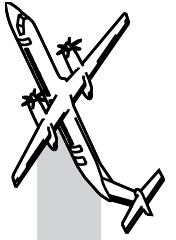


Figure 79-11. Oil Filler Cap and Tube Assembly



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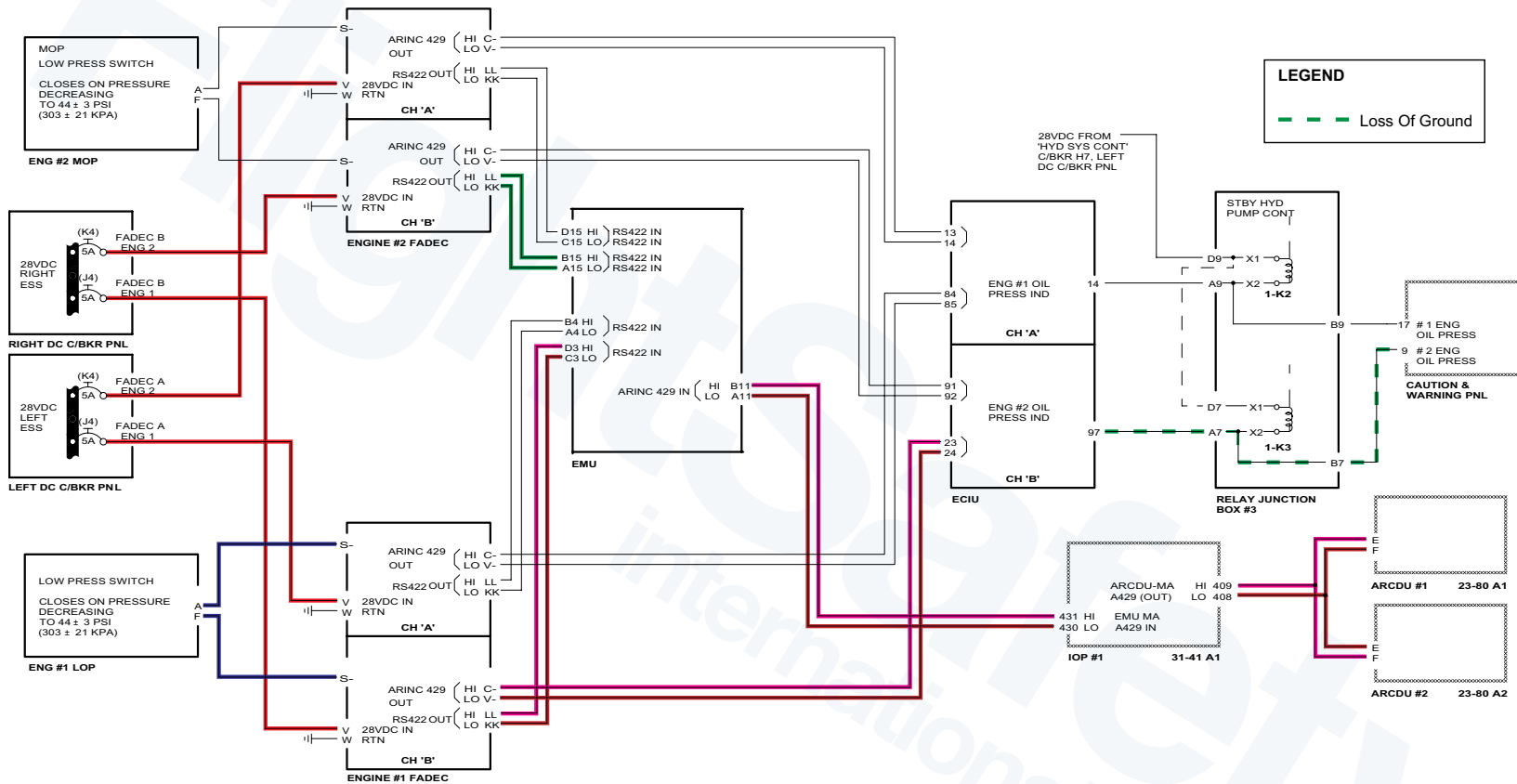
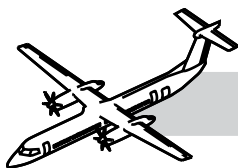


Figure 79-12. Low Oil Pressure No.1 Engine



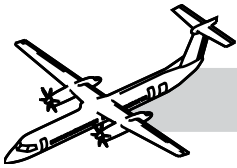
OPERATION

NOTE

Refer to Figure 79-12. Low Oil Pressure No.1 Engine.

During engine operation if the oil pressure drops to <44 psid, a signal will be sent from the low oil pressure switch to FADEC EEC:

- The signal will enter FADEC EEC of both channels
- Channel NOT in command would output data on ARINC 429 to the ECIU A and ECIU B
- ECIU B will remove the ground to bring ON the No.2 ENG OIL PRESS warning light
- FADEC EEC will output data on RS 422 bus to the EMU.
- EMU will output data for fault code display on ARCDU through the IOP.



79-30-00 OIL INDICATING SYSTEM

NOTE

GENERAL

The Oil Indicating System:

- Gives oil system status to the flight compartment
- Sends signals to the FADEC for temperature calculation
- Gives a visual oil level sighting during servicing.

SYSTEM DESCRIPTION

The Oil Indicating System uses switches, sensors and transducers to determine and transmit the engine oil system status to the flight compartment and the engine monitoring system.

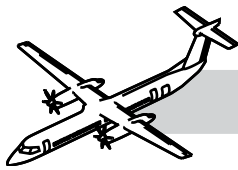
The engine oil is monitored for:

- Oil pressure
- Oil temperature and
- Ferrous chips.

Oil temperature is also used as a reference to adjust the indicated turbine temperature.

The Oil Indicating System includes the following sub-systems:

- Oil Pressure Indicating system
- Oil Temperature Indicating system
- Low Oil Pressure Warning system
- Chip Detection system
- Oil Filter Impending Bypass Warning system and
- Oil Quantity Indicating system.



79-31-00 OIL PRESSURE INDICATING SYSTEM

NOTE

GENERAL

Refer to Figure 79-13. Oil Pressure Indication Schematic.

The oil pressure indicating system has:

- An oil pressure transducer (to convert pressure to an electrical signal)
- A wiring harness (to transmit the pressure signal to the flight compartment).

The oil pressure transducer senses differential oil pressure at the outlet of the PRV.

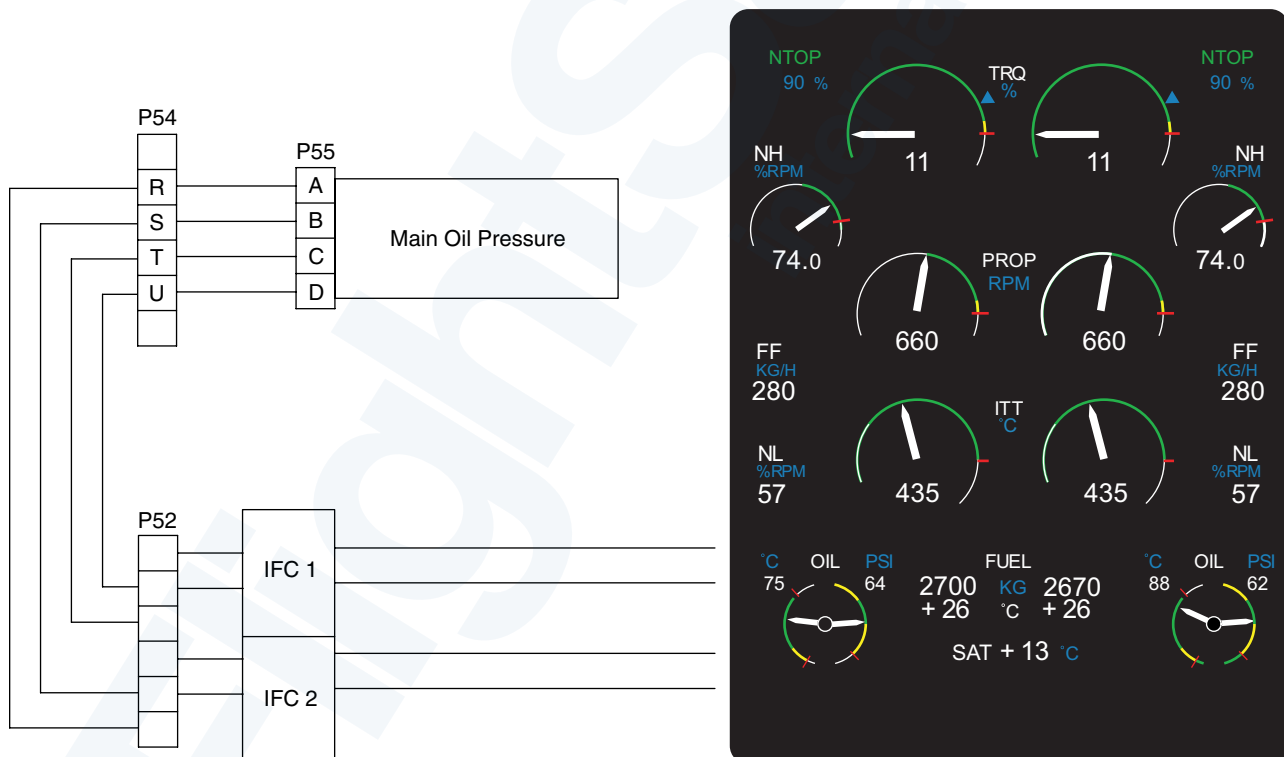
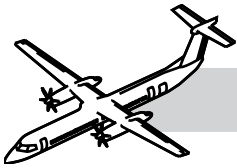


Figure 79-13. Oil Pressure Indication Schematic



COMPONENT DESCRIPTION

Oil Pressure Transducer

Refer to Figure 79-14. Oil Pressure Transducer.

The oil pressure transducer is at the bottom of the low pressure compressor case on the left side of the engine.

It is installed in an oil wetted cavity and it is held in place with two bolts and sealed with preformed pickings.

The oil pressure transducer sends an oil pressure differential signal to the IFC for display in the flight compartment.

Wiring harness connector (P55) is connected to the transducer.

A ground strap connects the housing of the transducer to the low pressure compressor case.

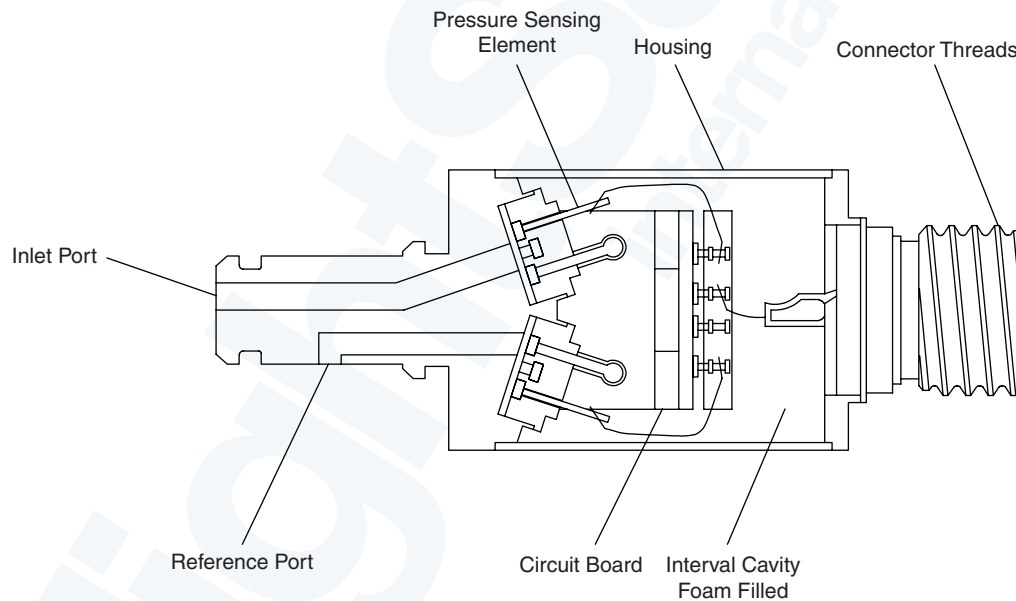
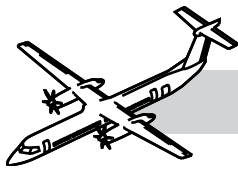


Figure 79-14. Oil Pressure Transducer



CONTROLS AND INDICATIONS

Refer to Figure 79-15. Oil Pressure and Temperature Indications.

Oil pressure and temperature are indicated on the ED in the flight compartment.

Pressure is indicated in PSI in both digital and analog formats.

Temperature is indicated in degrees C in both digital and analog formats.

For low oil pressure, an OIL PRESS warning light will come on. The EMU will make a record of the low oil pressure event when the:

- NH is more than 64%
- Condition lever has been in the START/FEATHER position for more than 20 seconds
- Switch has closed (due to low oil pressure).

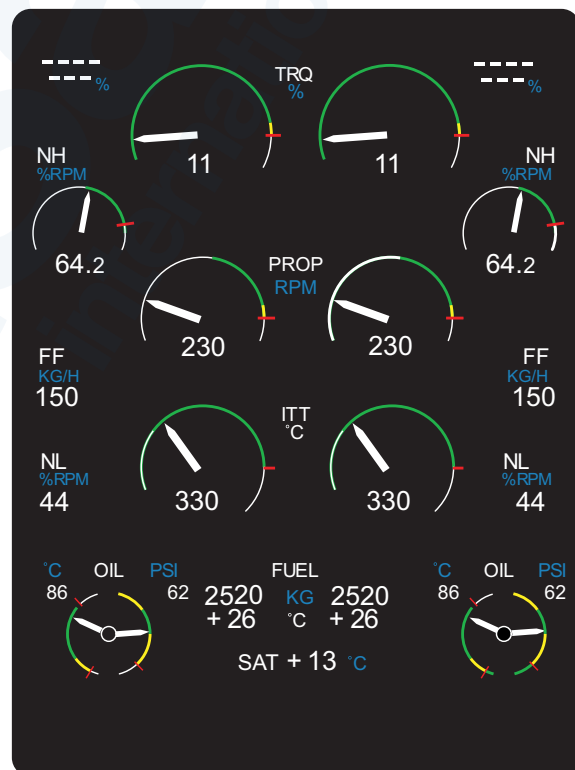
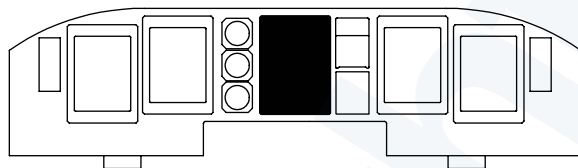


Figure 79-15. Oil Pressure and Temperature Indications

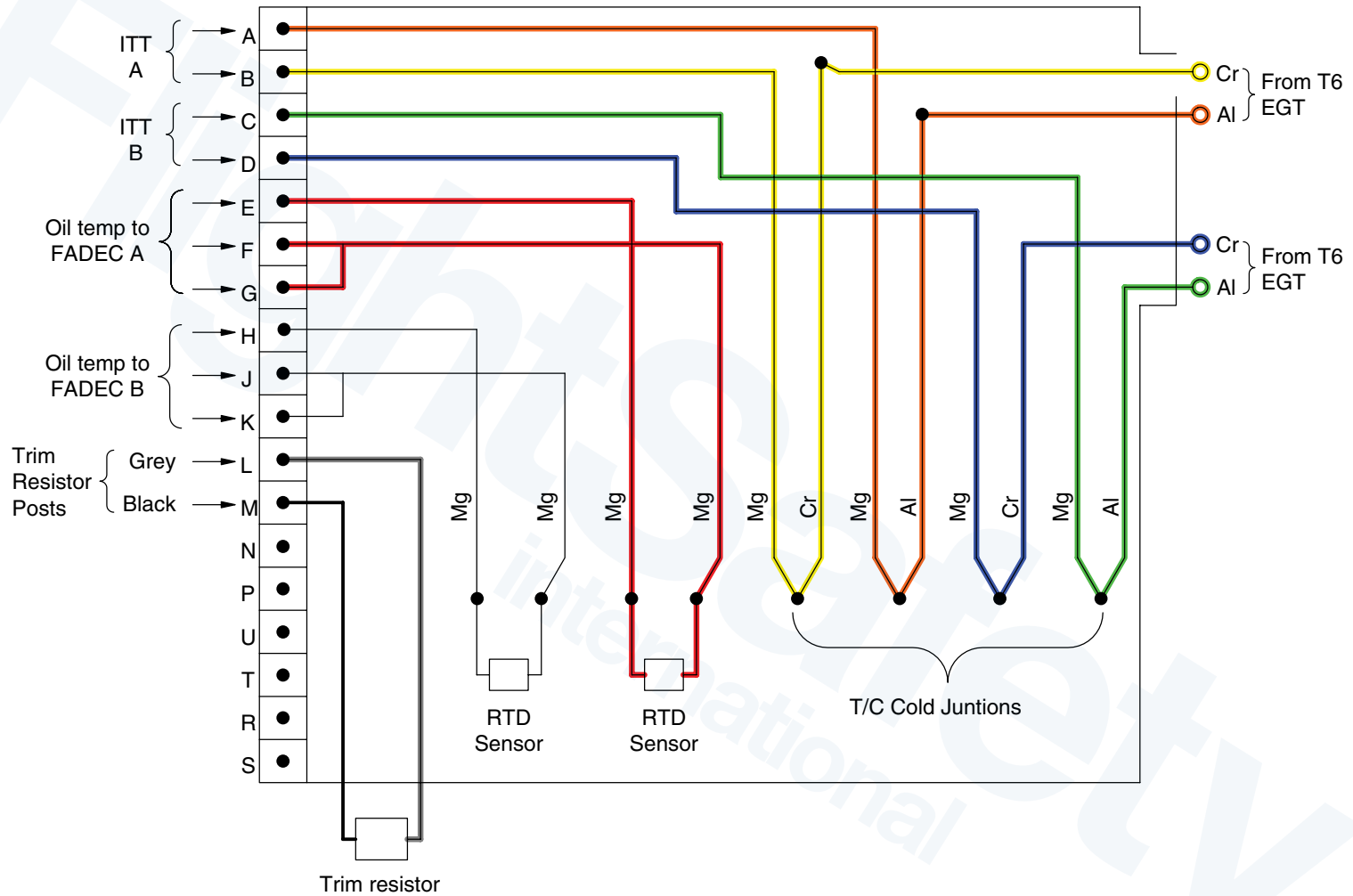
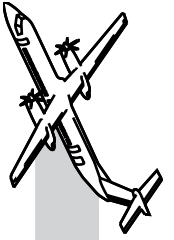
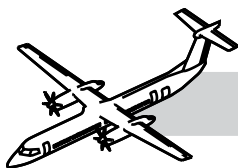


Figure 79-16. Oil Temperature Indicating System Schematic



79-32-00 OIL TEMPERATURE INDICATING SYSTEM

NOTES

Refer to Figure 79-16. Oil Temperature Indicating System Schematic.

The oil temperature indicating system has a two channel sensor and measures the main oil temperature.

Each channel provides an independent signal to the FADEC. The FADEC uses this signal as:

- A cold junction reference for the measurement and calculation of ITT
- An estimation of the torque shaft temperature
- Oil cooler door and ejector control
- An indication of main oil temperature to the flight compartment.

The sensor also provides a location to mount a resistor to trim the ITT signal to compensate for engine to engine differences in measured ITT.

The FADEC continually monitors the health of the MOT sensor and will generate a signal when the MOT signal is not in the limits.

FUNCTIONAL TEST OF THE MOT SENSOR

The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

- Do a visual check of the sensor
- Using an ohmmeter do a continuity resistance check between the receptacle pins and the terminals
- Using a megohmmeter do an insulation resistance check between the receptacle pins and the terminals.

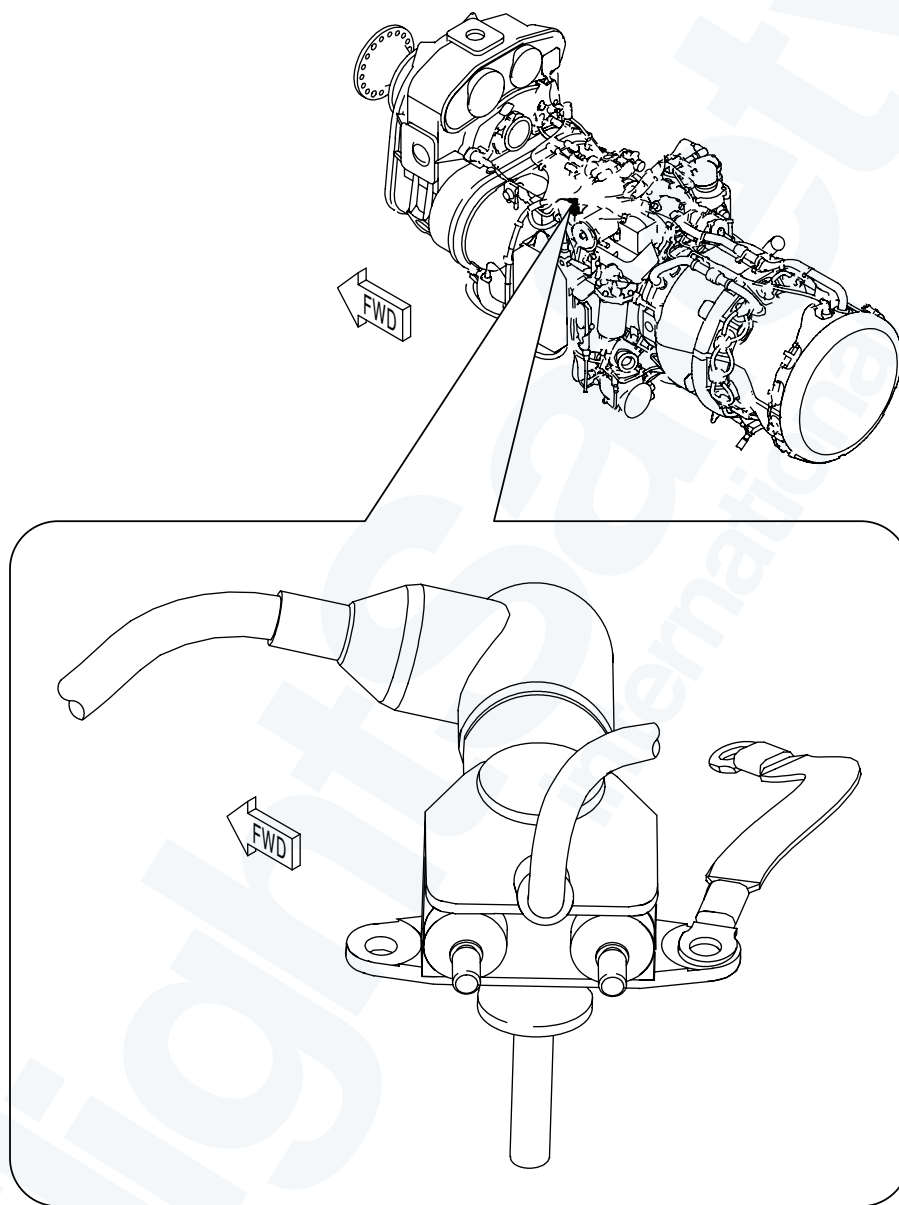
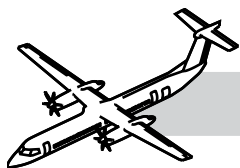
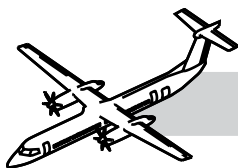


Figure 79-17. Oil Temperature Sensor - Location



COMPONENT DESCRIPTION

NOTE

Oil Temperature Sensor

Refer to:

- Figure 79-17. Oil Temperature Sensor - Location.
- Figure 79-18. Oil Temperature Sensor.

The oil temperature sensor, on the top of the air inlet case, sends an oil temperature signal to Channels A and B of FADEC.

It is installed in an oil wetted cavity and is held in place with two bolts and sealed with a preformed packing.

Wiring harness connector (P16) is connected to a receptacle on the sensor. The sensor signals are transmitted to the FADEC through the wiring harness.

A ground strap connects the housing of the sensor to the air inlet case.

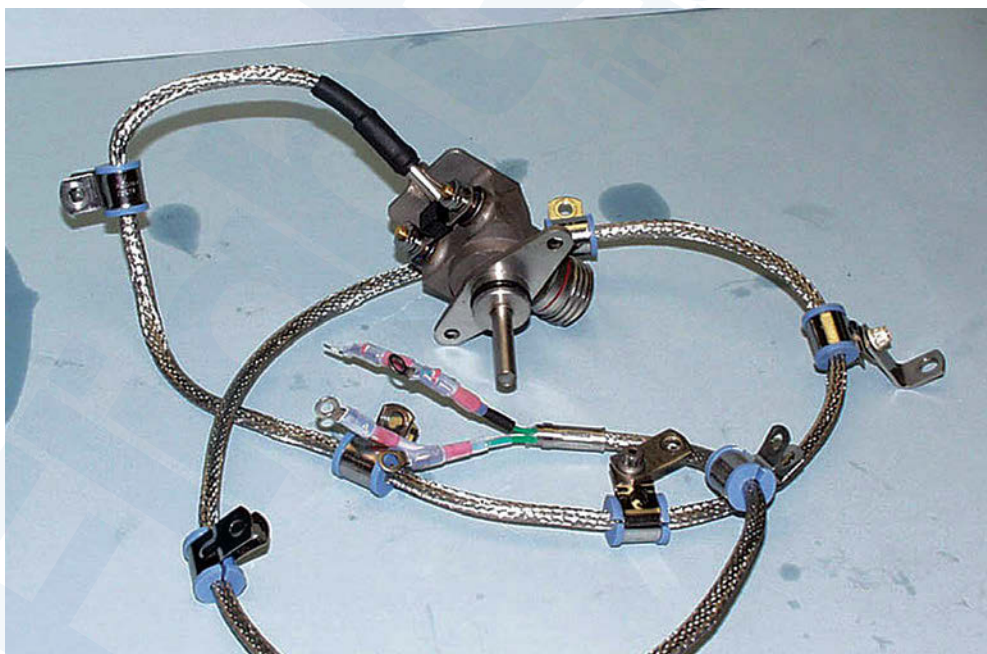


Figure 79-18. Oil Temperature Sensor

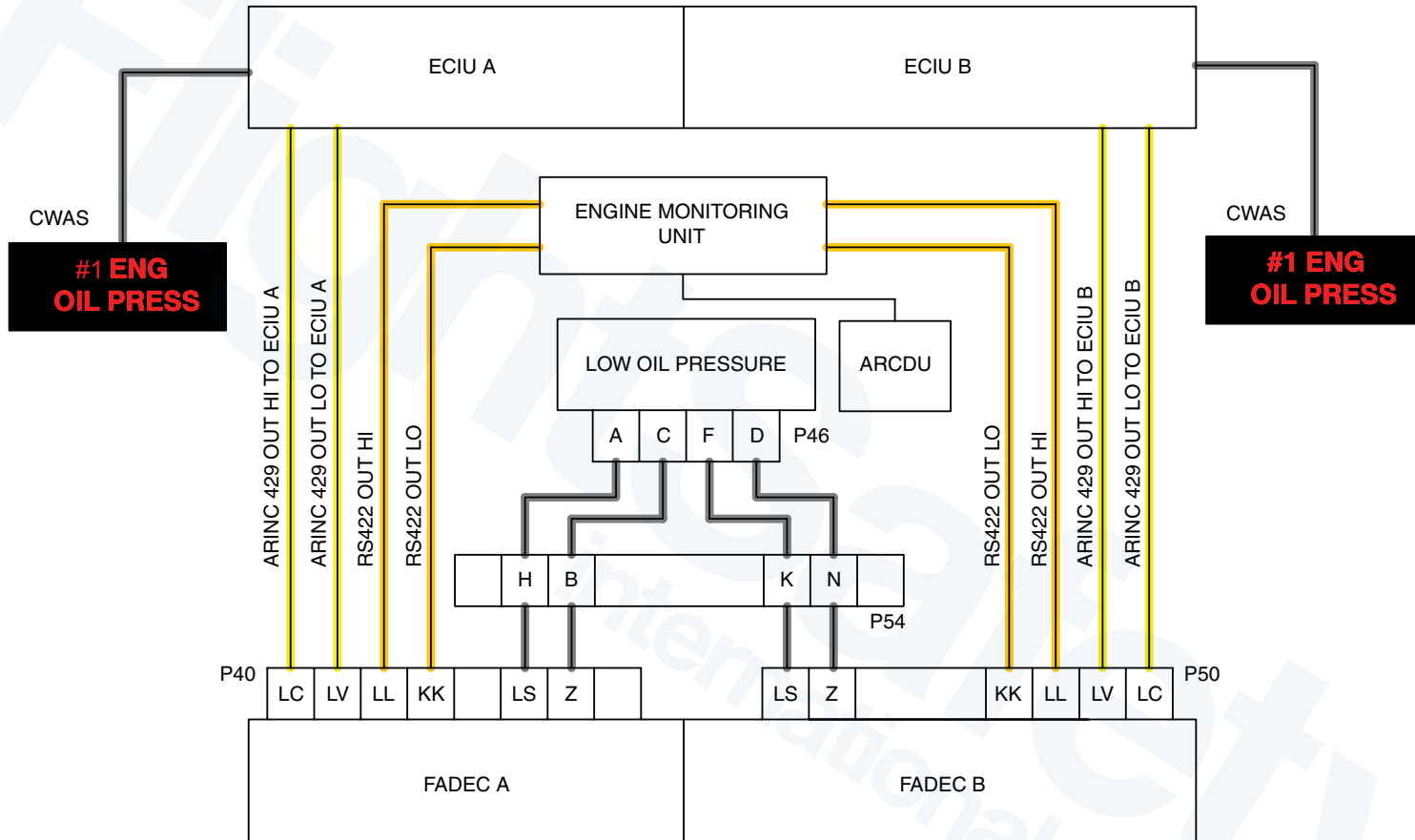
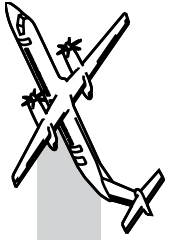
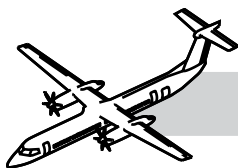


Figure 79-19. Low Oil Pressure Warning Schematic



79-33-00 LOW OIL PRESSURE WARNING SYSTEM

NOTE

SYSTEM DESCRIPTION

Refer to Figure 79-19. Low Oil Pressure Warning Schematic.

The No.1 ENG OIL PRESS warning light comes on when the (FADEC1) senses any condition that follows:

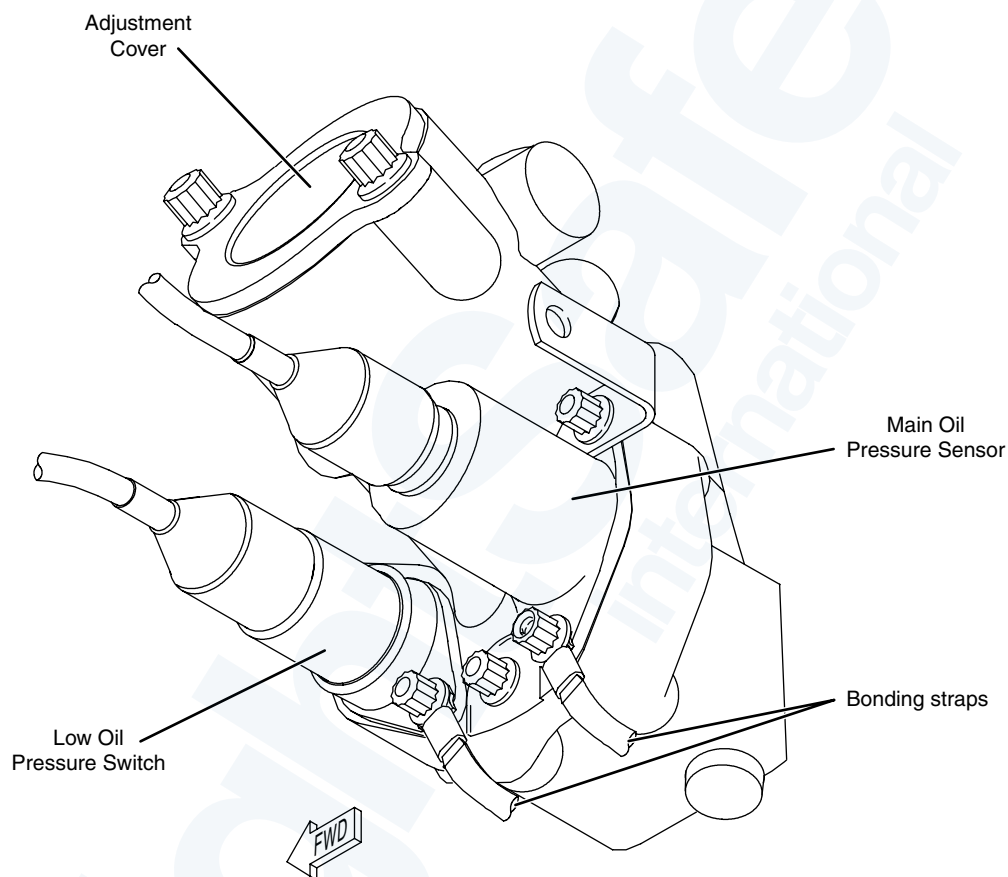
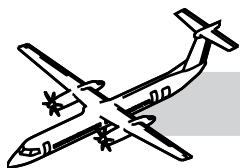
- Main oil pressure is less than 44 psi
- Main oil pressure crosscheck latched failed for more than 15 seconds.

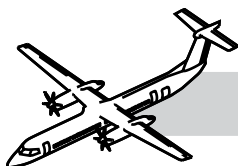
NOTE

A low oil pressure switch fault code is set when the NH is more than 64% and the main oil pressure is less than 44 psi.

NOTE

The FADEC supplies discrete data through the ECIU to the CAWP for the No.1 ENG OIL PRESS warning indication.

**Figure 79-20. Low Oil Pressure Switch - Detail**



COMPONENT DESCRIPTION

NOTE

Low Oil Pressure Switch

Refer to:

- Figure 79-20. Low Oil Pressure Switch - Detail.
- Figure 79-21. Low Oil Pressure Switch.

The low oil pressure switch is at the bottom of the low pressure compressor case on the left side of the engine. It detects low oil pressure and sends a signal to the FADEC.

It is installed in an oil wetted cavity and is held in place with two bolts. It is sealed with preformed packings.

The wiring harness connector is connected to the switch. This connector transmits the switch signal to the FADEC through the wiring harness.

A ground strap connects the housing of the switch to the low pressure compressor case.



Figure 79-21. Low Oil Pressure Switch

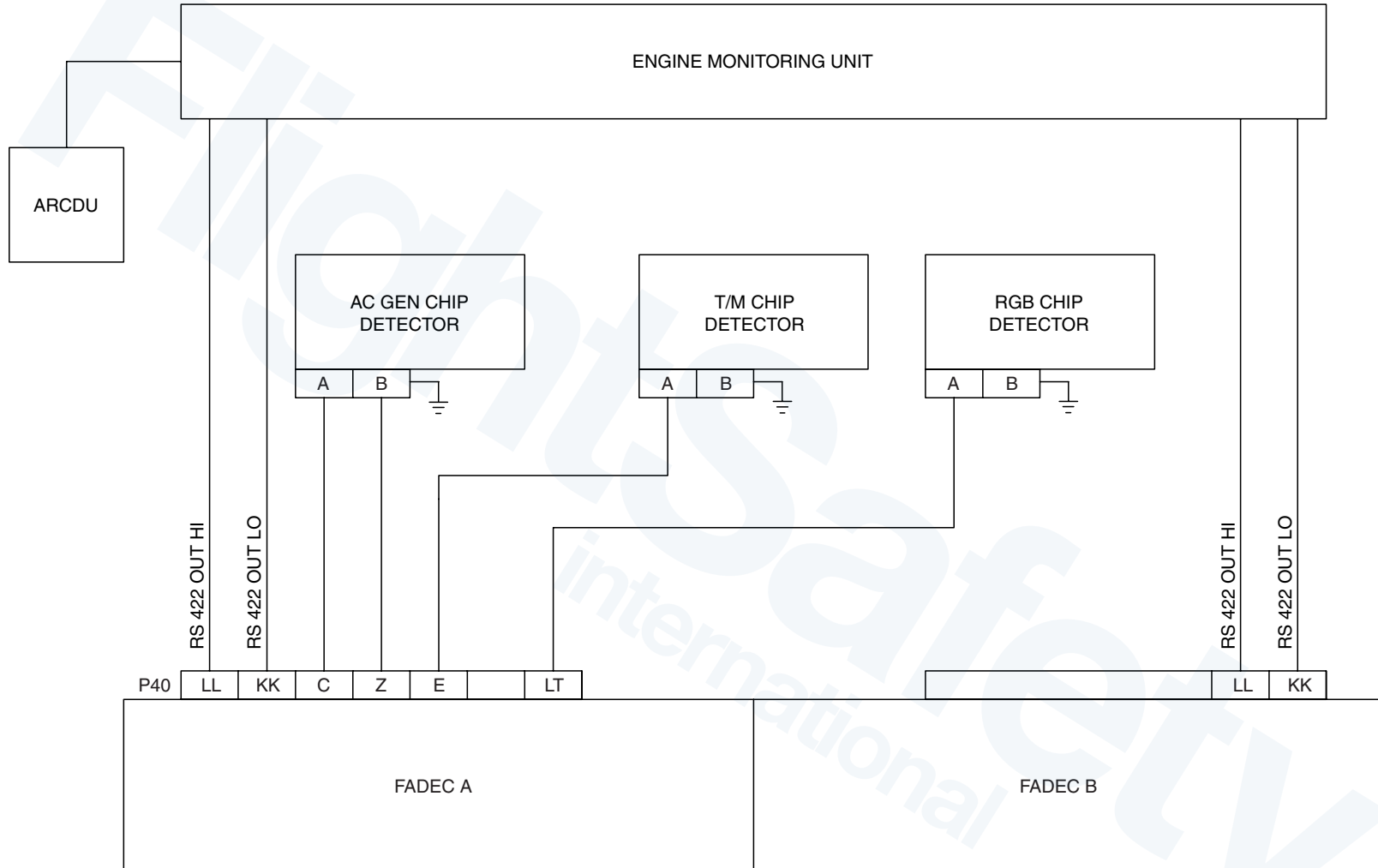
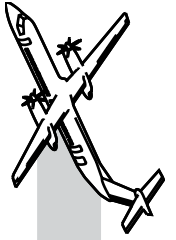
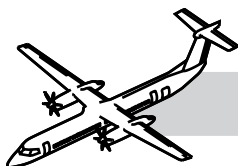


Figure 79-22. Chip Detection System - Schematic



79-34-00 CHIP DETECTION SYSTEM

NOTE

GENERAL

Refer to Figure 79-22. Chip Detection System - Schematic.

The chip detection system uses three chip detectors to indicate the presence of ferrous chips coming from the:

- AC generator
- Turbomachinery module
- RGB module.

Chip detector fault codes are as follows:

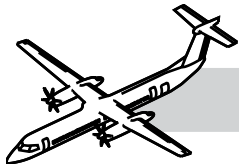
- 937 AC Generator
- 938 Turbomachinery
- 939 Reduction Gearbox

SYSTEM DESCRIPTION

The system signals are transmitted electrically to the FADEC channel A and transferred internally to FADEC channel B.

Ferrous debris in the oil system is attracted to the chip detector poles.

If the poles are bridged (by debris), a signal is sent to the FADEC. The FADEC will then output a RS422 signal to the EMU.



DASH 8 Q400 MAINTENANCE TRAINING MANUAL

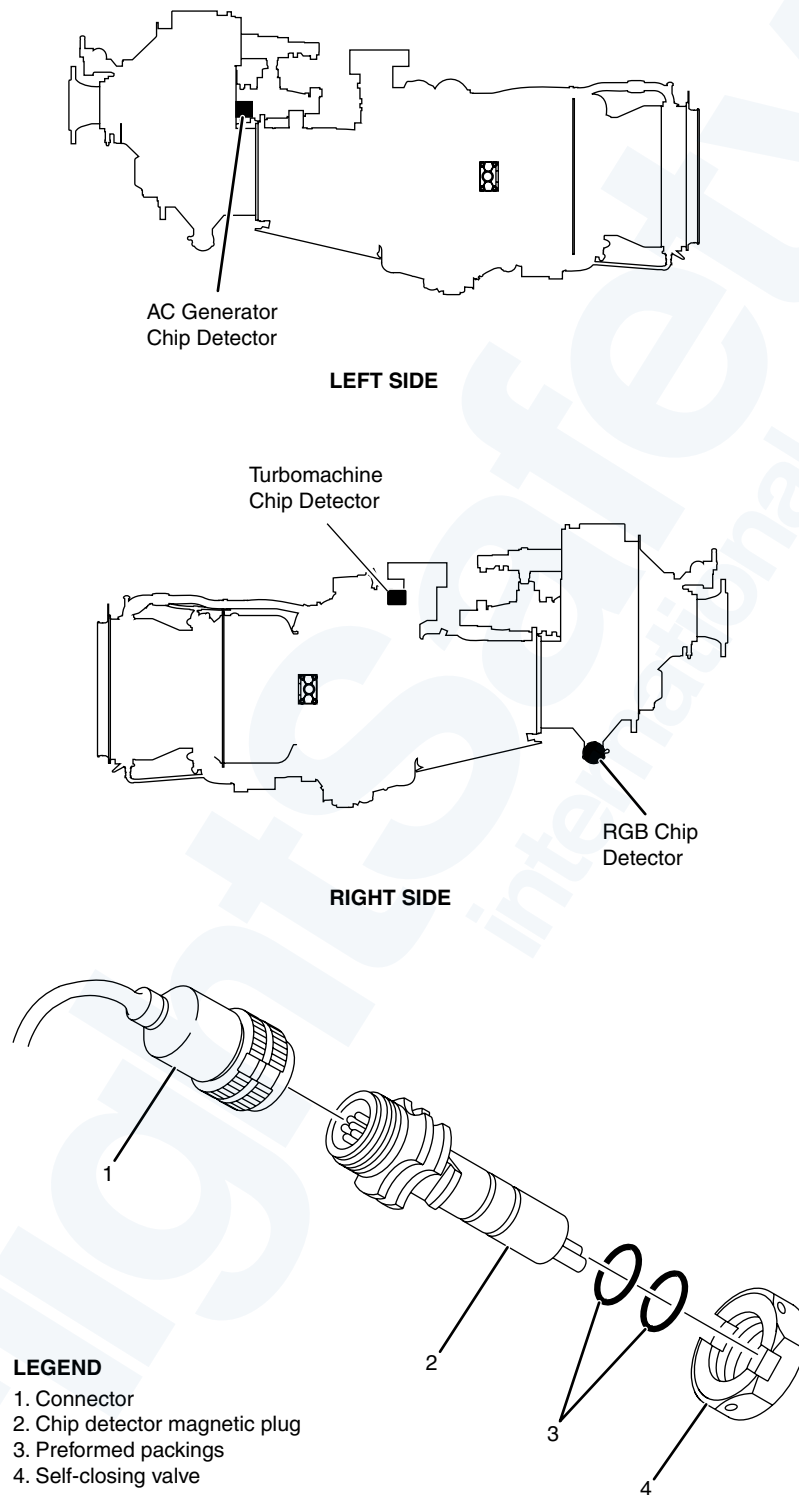
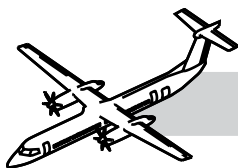


Figure 79-23. Chip Detector Magnetic Plug



VISUAL CHECK OF THE CHIP DETECTOR INDICATING SYSTEM

NOTE

The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

Refer to Figure 79-23. Chip Detector Magnetic Plug.

Can be checked in two methods as follows:

Using ARCDU:

1. Set Other Systems
2. Set EMU
3. Set Event History
4. Set Engine Health History
5. Set Chip Detector
6. Look at the display and find the chip detector that gives the indication
7. Visual inspect the chip detector that generated the fault
8. If the RGB detector gave the indication and is contaminated, a borescope inspection of the RGB is mandatory.

Using the Engine Display:

1. PLA to Flight Idle
2. CLA to Fuel Off
3. Maintenance discrete switch selected
4. Push the MCL switch
5. Fault codes 937, 938 or 939 displayed, chip has been detected.

For both methods consult FIM if a chip is indicated.

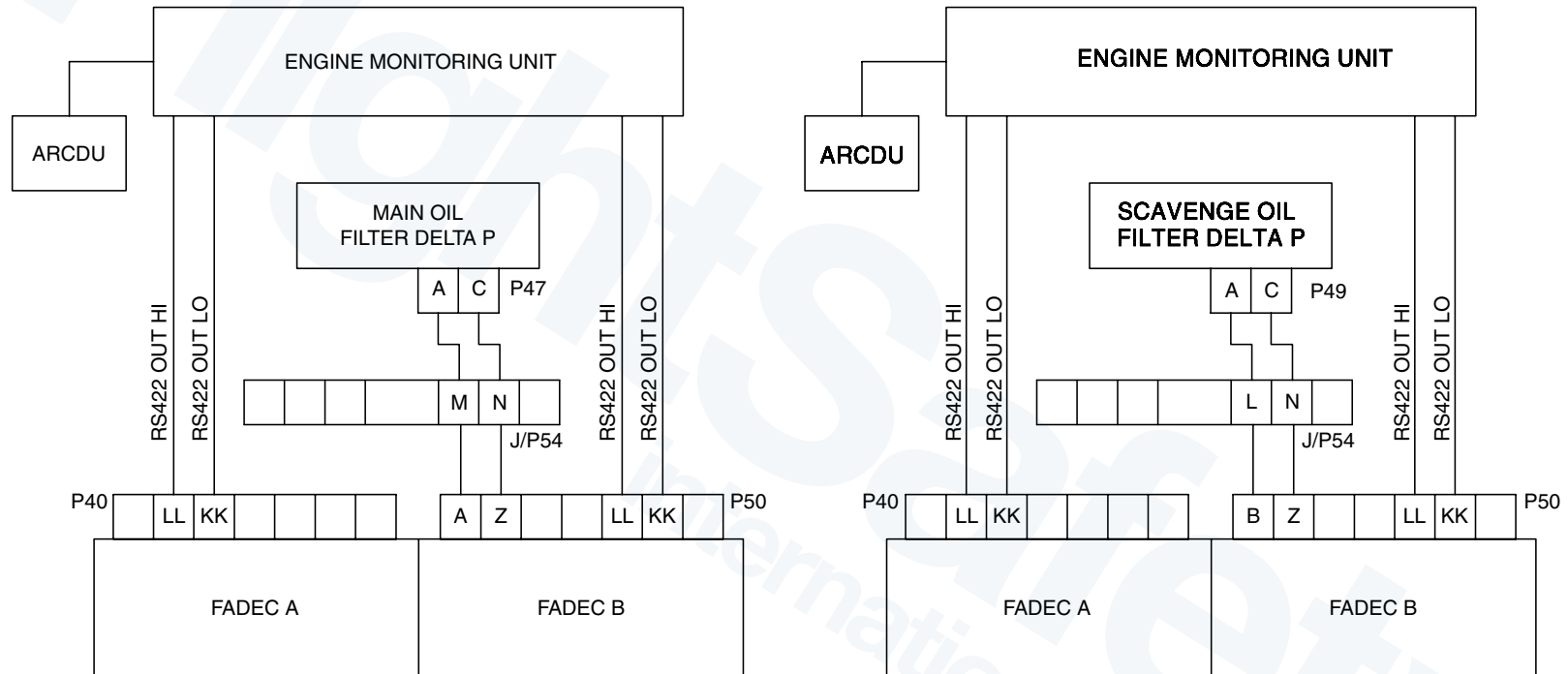
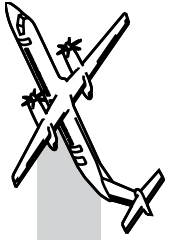
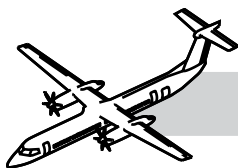


Figure 79-24. Oil Filter Impending Bypass Warning System - Schematic



79-35-00 OIL FILTER BYPASS WARNING SYSTEM

NOTE

GENERAL

Refer to Figure 79-24. Oil Filter Impending Bypass Warning System - Schematic.

The oil filter impending bypass warning system warns of an impending bypass of oil across the oil filter. This indicates a contaminated oil filter.

SYSTEM DESCRIPTION

When the oil filter differential pressure exceeds 22 psid (152 kPad), the pressure switch opens.

This state change is detected by the FADEC, which in turn sends a discrete signal to the EMU.

The EMU makes a record of a main oil filter impending bypass event.

During operation with cold oil (main oil temperature less than 175°F (80°C)), any impending bypass signal is ignored by the FADEC.

This will prevent false event messages from being stored in the EMU.

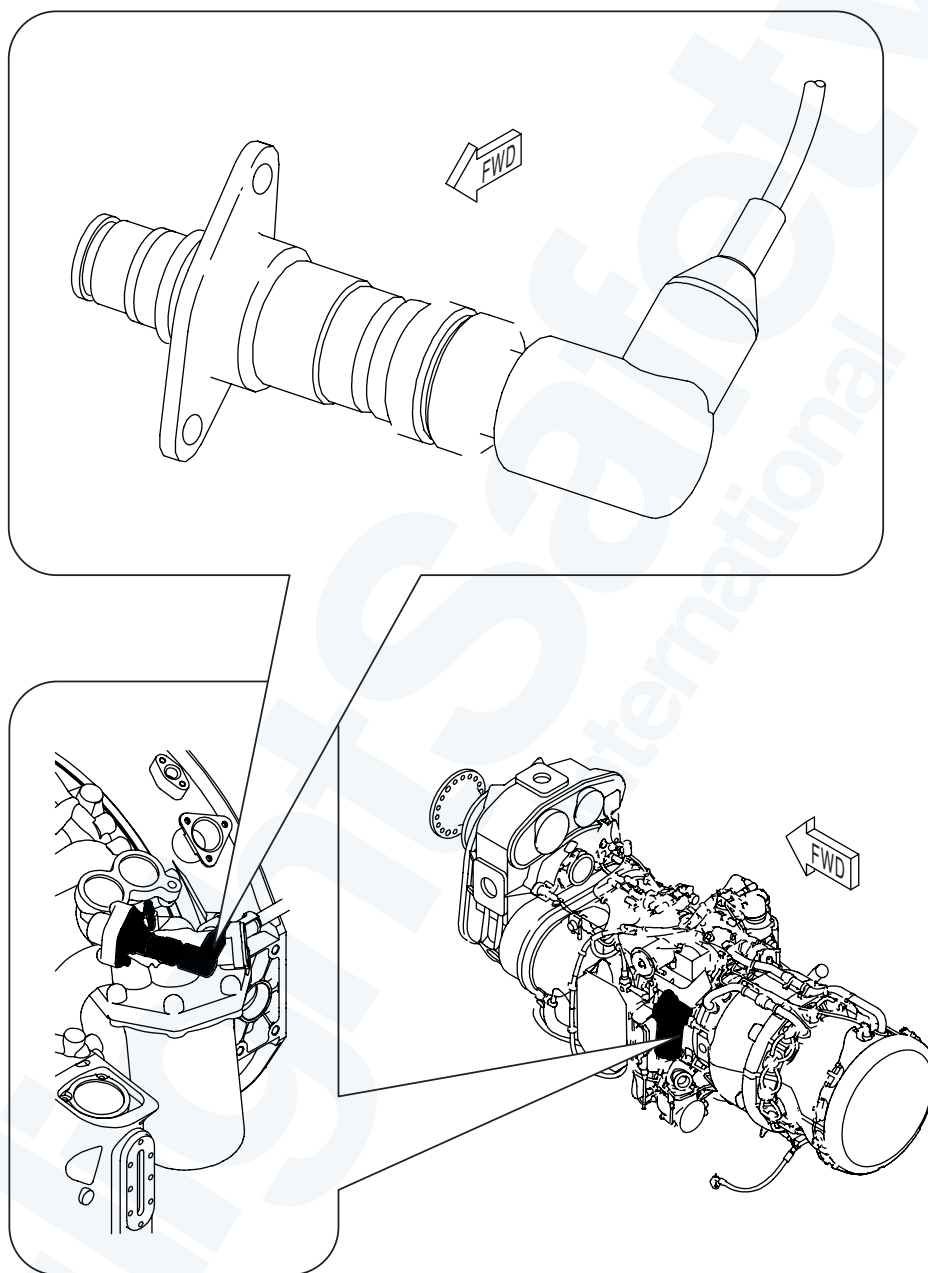
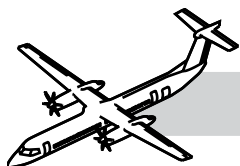
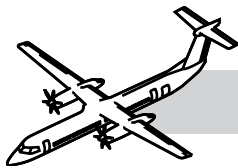


Figure 79-25. Oil Filter Impending Bypass Switch



COMPONENT DESCRIPTION

NOTE

Oil Filter Impending Bypass Switch

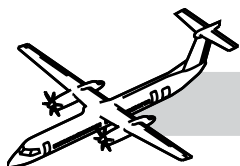
Refer to Figure 79-25. Oil Filter Impending Bypass Switch.

There are two impending bypass switches in the system.

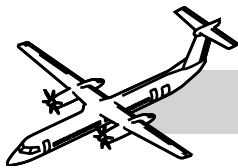
One is on the main oil filter housing; the other on the scavenge oil filter housing. They are installed in oil wetted cavities that are attached with bolts and sealed with preformed packings.

The oil filter impending bypass switch detects the pressure drop across a partially blocked engine oil filter and sends a signal to FADEC.

Wiring harness connectors (P47 and P49) connect to the switches. The connectors transmit the switch signals to the FADEC channel B which sends it internally to channel A. Ground straps connect the switch housings of the transducer to the engine cases.



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79-21-00 OIL COOLER SYSTEM

NOTE

GENERAL

The oil cooler system removes excess heat from the oil to keep the temperature within limits. This ensures that the oil retains its viscosity.

SYSTEM DESCRIPTION

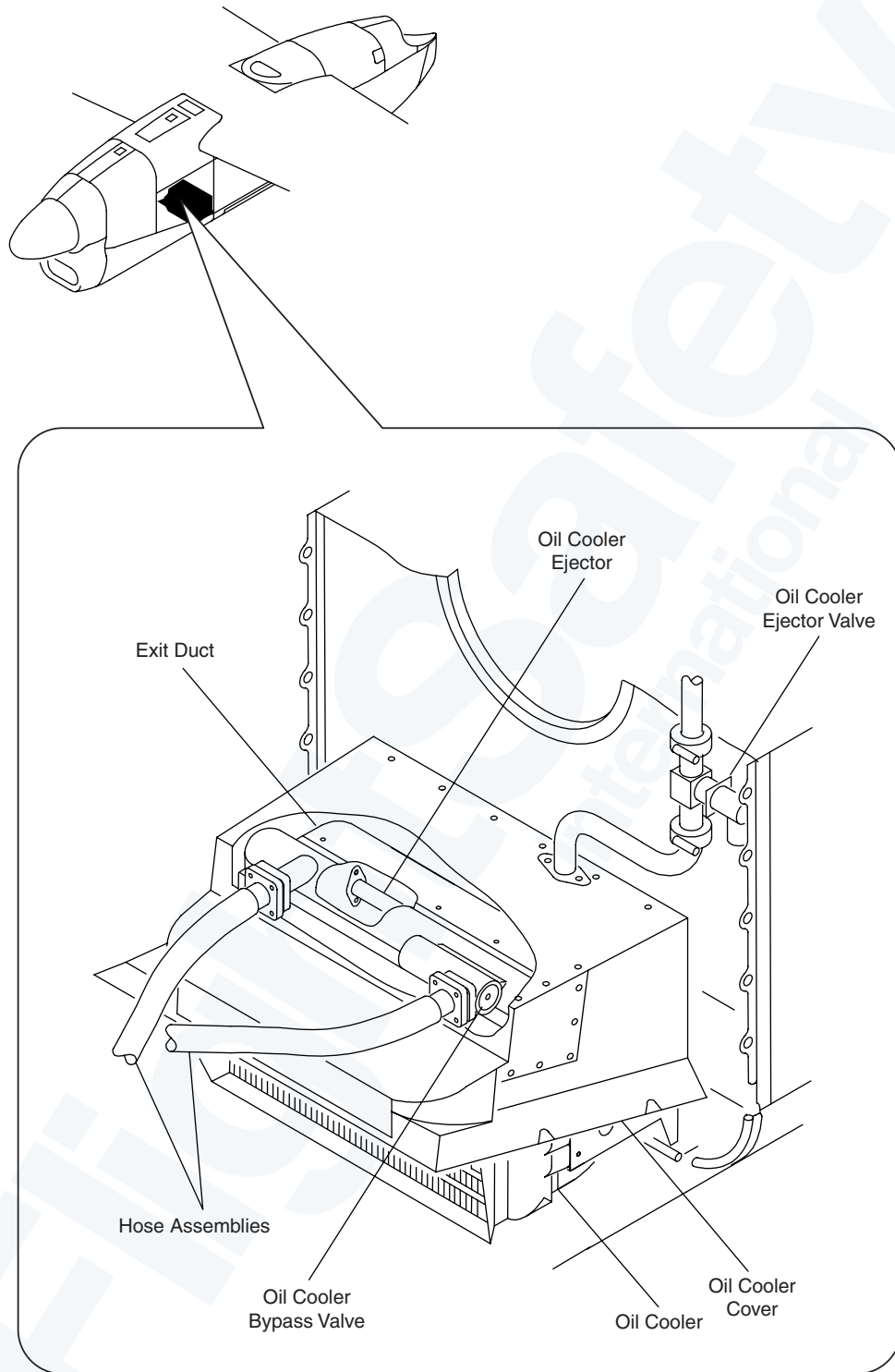
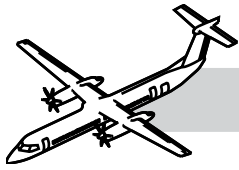
The oil cooler is in the oil pressure circuit of the engines. The oil cooler system contains the following components:

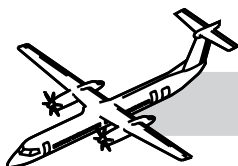
- Oil cooler
- Oil cooler bypass valve
- Exit duct
- Oil cooler ejector
- Oil cooler ejector valve
- Oil cooler air outlet flap and
- Oil cooler air outlet flap actuator.

The oil cooler uses the air from the engine air intake to cool the oil.

The air inlet duct is welded to the oil cooler. It directs airflow through the oil cooler.

The exit duct is not part of the assembly. It is attached to the oil cooler with bolts and anchor nuts.

**Figure 79-26. Oil Cooler**



COMPONENT DESCRIPTION

Oil Cooler

The oil cooler is attached between two brackets at the engine firewall structure.

Refer to:

- Figure 79-26. Oil Cooler.
- Figure 79-27. Oil Cooler Side View.
- Figure 79-28. Oil Cooler Bottom View.

The oil cooler unit is in a cutout in the lower cowl and is supplied with air from the inlet duct.

It is enclosed in a titanium shroud with fireproof seals against the cowl to isolate it from the fire zone.

A drain plug is installed at the lowest point on the oil cooler.

Oil Cooler Bypass Valve

The oil cooler bypass valve is installed in the left side of the oil cooler. It performs the functions listed below:

A thermal function lets the engine oil fully bypass the oil cooler at temperatures below 170°F (77°C). Above 185°F (85°C) the engine oil passes through the oil cooler.

A pressure relief function lets the engine oil bypass if the oil cooler becomes blocked during cold oil operation. The pressure relief operates at 23 psid (158.6 kPad).



Figure 79-27. Oil Cooler Side View

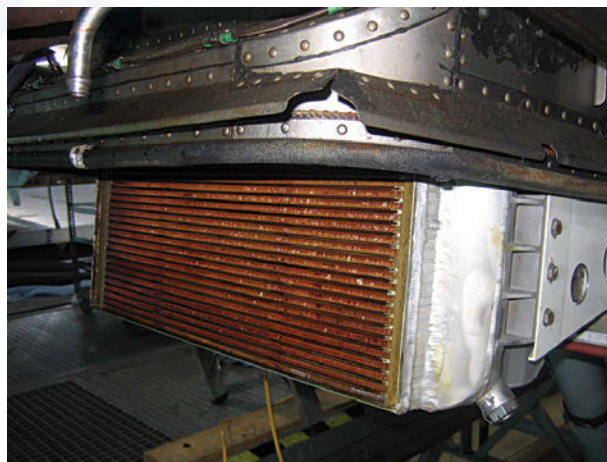
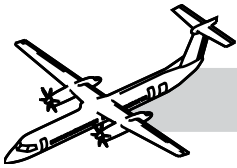


Figure 79-28. Oil Cooler Bottom View



Oil Cooler Ejector

NOTE

Refer to Figure 79-29. Oil Cooler Ejector.

The oil cooler ejector is installed in the exit duct, aft of the oil cooler.

The oil cooler ejector operates when the aircraft is on the ground. It uses bleed air from downstream of the precooler to eject at high pressure through the 22 holes. This causes air in the nacelle to be pulled through the oil cooler heat-exchanger.

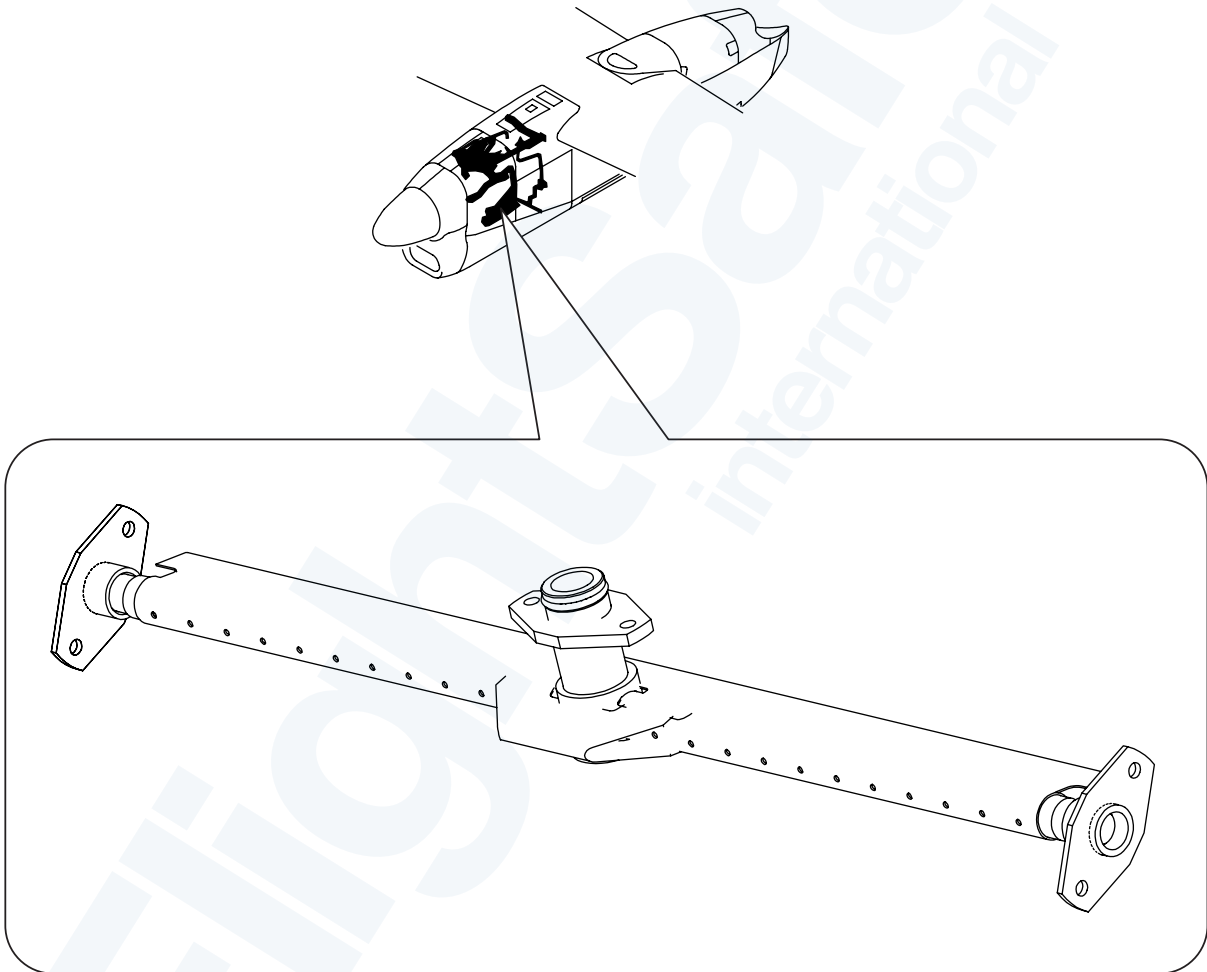
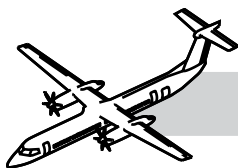


Figure 79-29. Oil Cooler Ejector



Oil Cooler Ejector Valve

NOTE

Refer to Figure 79-30. Oil Cooler Ejector Valve.

The oil cooler ejector valve is on the forward face of the engine firewall and controls the flow of air to the ejector.

An energized solenoid shuts the valve when the aircraft is airborne.

Air supply is from the bleed air system before the Nacelle Shut-Off Valve (NSOV).

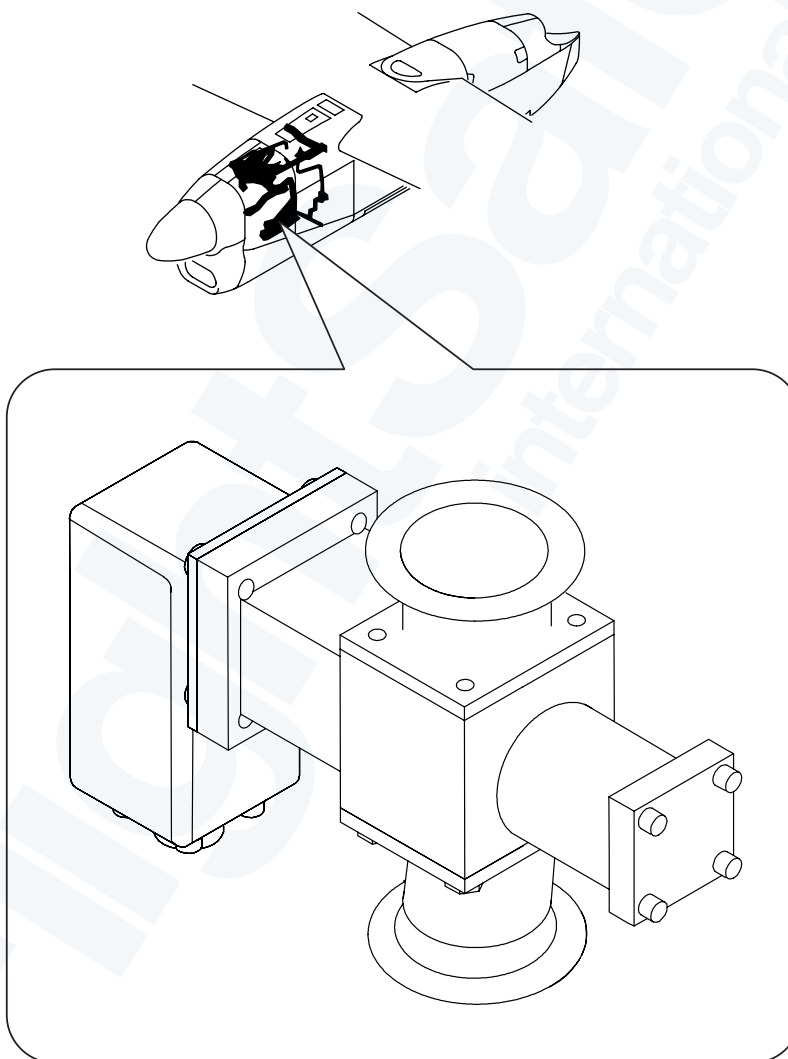


Figure 79-30. Oil Cooler Ejector Valve

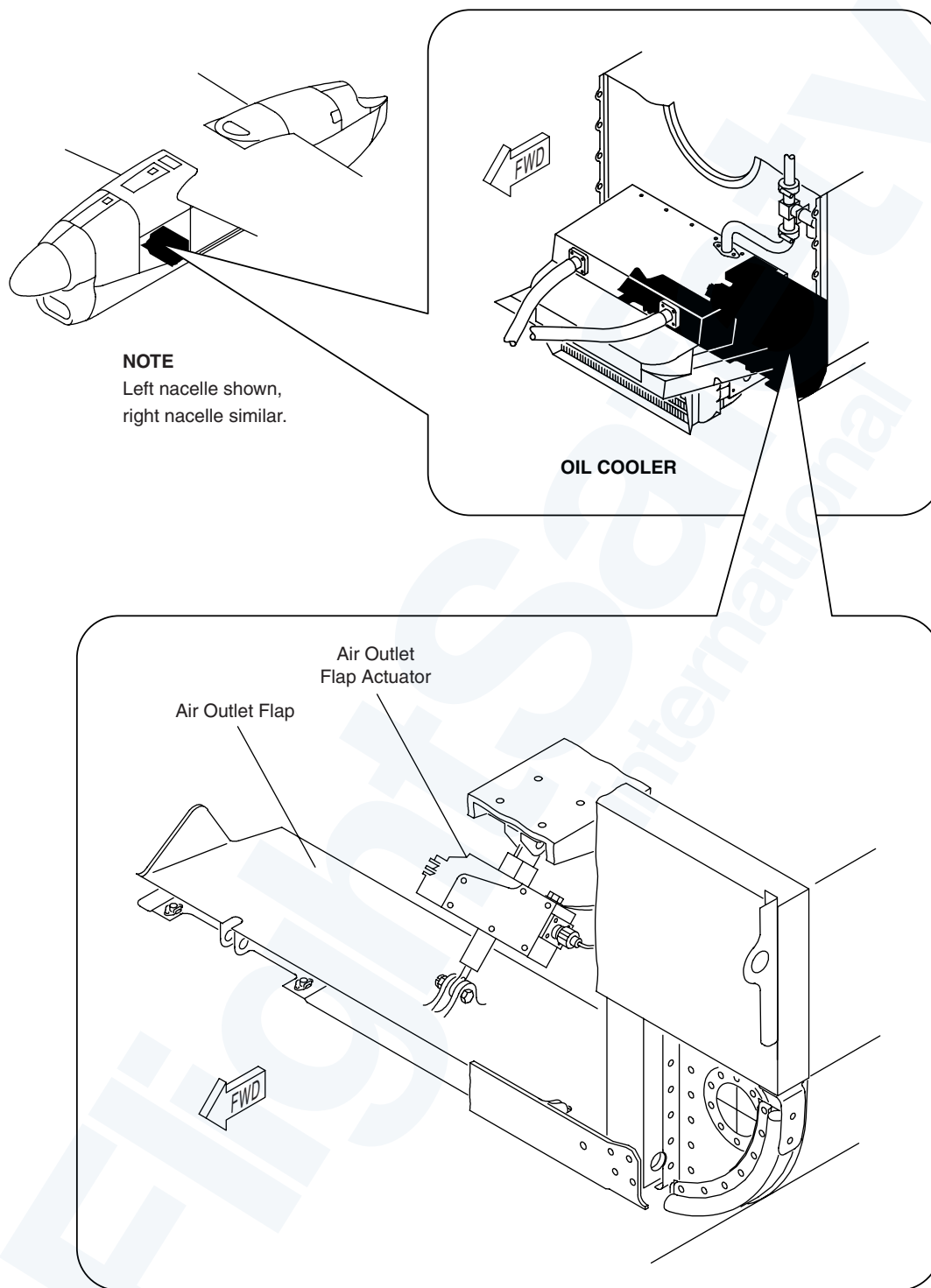
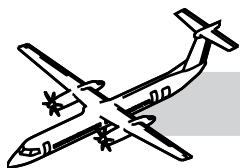
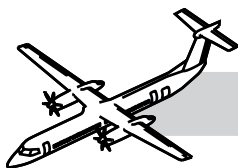


Figure 79-31. Oil Cooler Air Outlet Flap



Oil Cooler Air Outlet Flap

NOTE

Refer to Figure 79-31. Oil Cooler Air Outlet Flap.

The oil cooler air outlet flap is in the bottom surface of the nacelle, FWD of the engine firewall. It is hinged at its forward end and is operated by an actuator.

This flap controls the flow of air through the oil cooler and is fully closed when the engine oil bypasses the oil cooler.

When it is necessary for the engine oil to be cooled, the flap is open in one of three positions. This lets the necessary airflow exit to the atmosphere.

When the oil temperature is more than 102°C and airspeed < 170 KIAS cooler, the flap is fully open at 32°.

When the oil temperature is between 85°C and 95°C, the flap is open at 5°.

When the engine oil temperature is between 98°C and 89°C, the flap opens at 10°.

It is fully closed when the oil temperature is less than the limit on the ground.

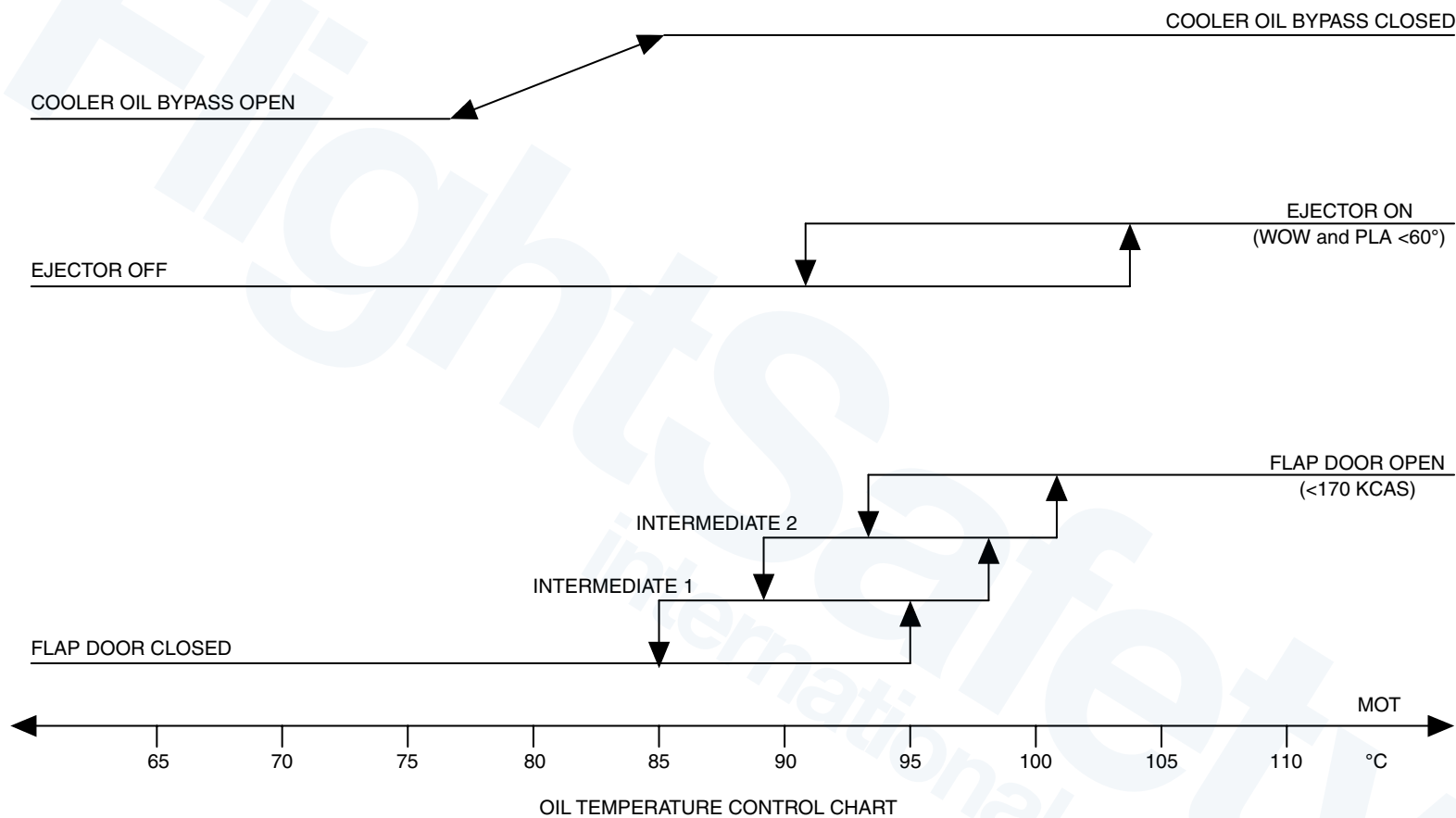
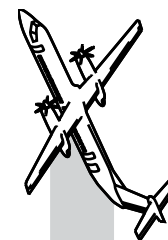
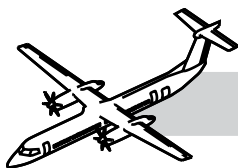


Figure 79-32. Oil Temperature Chart



Air-Cooled Oil Cooler Air Outlet Flap Actuator

Refer to Figure 79-32. Oil Temperature Chart.

The oil cooler air outlet flap actuator is to position the outlet flap to the position commanded by FADEC.

The oil cooler air outlet flap actuator is attached to the nacelle structure and the oil cooler air outlet flap.

It receives signals from the FADEC, through the ECIU.

Test

Test the operation of the air-cooled oil cooler flap actuator using this procedure:

First, to do actuator operational test, it is necessary to do a check of the travel of the ACOC air outlet flap.

To perform the ACOC air outlet flap check:

- In the wardrobe, on the CENTRAL maintenance panel, set CDS GND MAINT switch
 - In the flight compartment, on the aft center console, set either ARCDU selector to the ON position
 - Push the MAINT key on the ARCDU
 - Select the menu items that follow:
 - POWERPLANT MAIN
 - POWERPLANT INTERFACE
 - ACOC FLAP DOOR AND EJECTOR STATUS.
 - Move the Power Lever Assembly to RATING detent
- Make sure that both Condition Levers are at FUEL OFF
 - Set the MAINT DISC switch to the ON position to simulate a Main Oil Temperature (MOT) of 75°C. Make sure that:
 - The ED on the PFD shows an MOT of 75°C
 - The outlet flap is fully closed.
 - On the Engine Control Panel, push the RDC TOP RESET button to simulate an MOT of 107°C. Make sure that:
 - The ED shows an MOT of 107°C
 - The outlet flap moves smoothly to the fully open position.
 - Press the RDC TOP DEC button on the Engine Control Panel again and again.

NOTE

Each time the button is pressed, simulates a fall in MOT of 2°C.

Make sure that the air outlet flap closes in increments as the oil temperature decreases through 93°C, 89°C, and 85°C. Make sure that the travel is smooth.

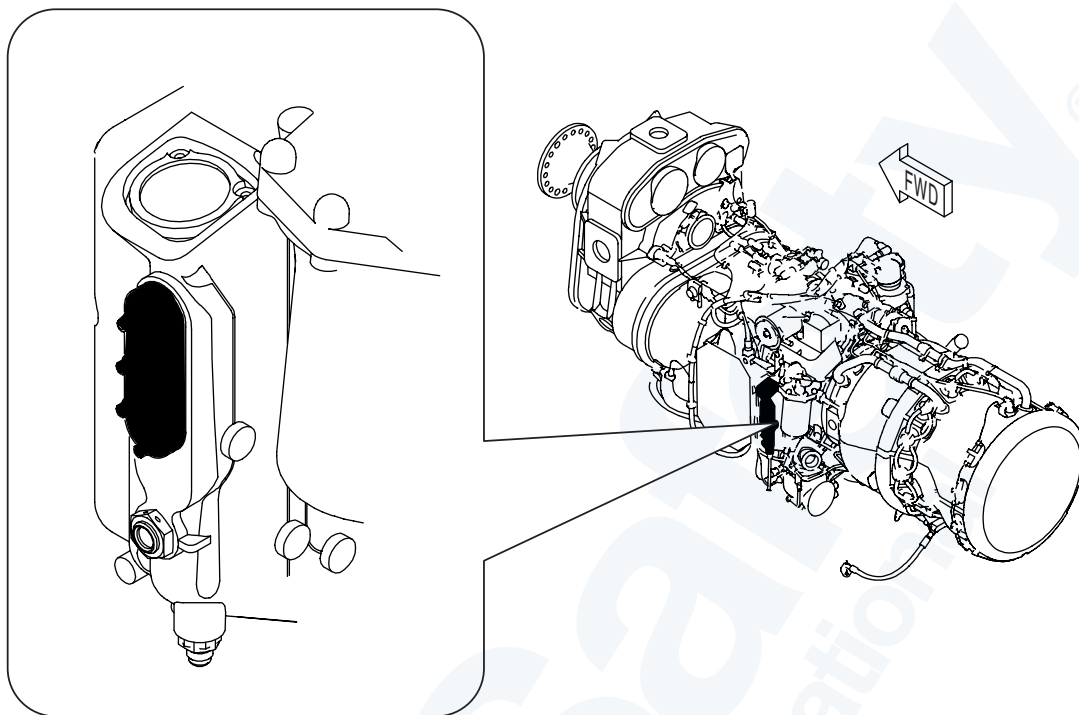
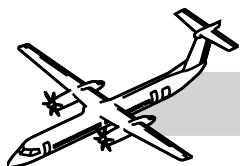


Figure 79-33. Oil Level Indicator Sight Glass - Location

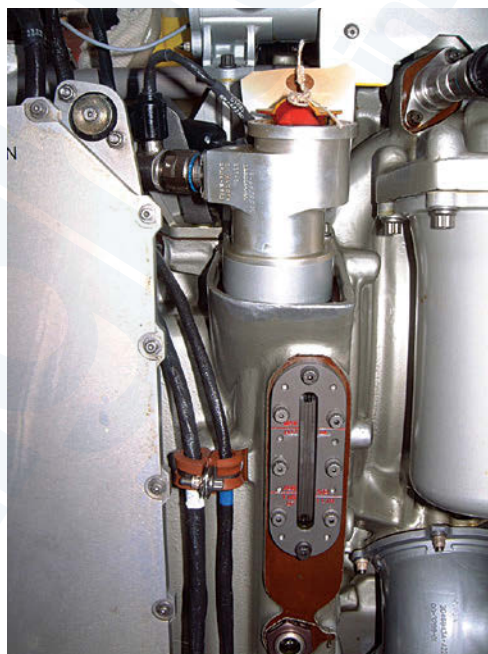
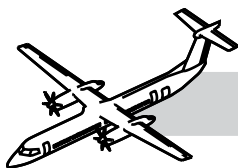


Figure 79-34. Oil Level Indicator Sight Glass



79-36-00 OIL QUANTITY INDICATING SYSTEM

GENERAL

The low oil level indicator is used to give a quick visual oil quantity indication in the tank.

The oil quantity indicating system contains the following components:

- Oil level indicator sight glass
- Calibrated dipstick and a
- Low oil level indicator glass to indicate oil quantity.

The sight glass and the dipstick are used during normal engine servicing.

COMPONENT DESCRIPTION

Oil Level Indicator Sight Glass

Refer to:

- Figure 79-33. Oil Level Indicator Sight Glass - Location.
- Figure 79-34. Oil Level Indicator Sight Glass.

The oil level indicator sight glass gives a further method to visually check the oil level in the tank.

The oil level indicator sight glass is on the oil tank on the left side of the engine.

The sight glass is used during normal engine servicing. It gives a visual indication of the oil level in the tank and has markings to indicate the quantity of oil to add (if required).

The sight glass is attached to the oil tank with 8 bolts and is sealed with preformed packing.

CHECK OF THE ENGINE OIL LEVEL AND REPLENISH AS NECESSARY

The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

The maintenance procedure that follows is to check the engine oil level and replenish as necessary.

WARNING

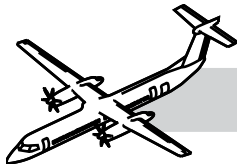
BE CAREFUL WHEN YOU DO WORK ON THE ENGINE COMPONENTS IMMEDIATELY AFTER THE ENGINE IS STOPPED. THE ENGINE COMPONENTS CAN STAY HOT FOR ONE HOUR AND CAN CAUSE INJURY.

WARNING

MAKE SURE THAT THE OIL IS NOT HOT WHEN YOU DO MAINTENANCE. THE HOT OIL CAN BURN YOU.

WARNING

DO NOT LET THE OIL TOUCH YOUR SKIN FOR A LONG TIME. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.



Do the engine oil level check as follows:

1. Locate the engine oil level sightglass
2. Clean the sightglass with the lint-free cloth.
3. Add oil as necessary. If the oil level is between the MAX HOT indication and the ADD 1 LTR indication lines, it is not necessary to add oil
4. If the oil level is at or below the ADD 1 LTR indication line, it is necessary to add oil

NOTE

You must examine the oil level between 15 minutes and 30 minutes after engine shutdown. If you do not, the oil level sightglass indication will not be accurate.

5. Do an engine run to circulate the oil if the aircraft has been parked overnight

Run the engine for a minimum of 30 seconds with the oil temperature in the green section of the oil temperature gauge and propeller in feather during the shutdown

This will make sure that the maximum amount of oil returns to the oil tank from the propeller control unit.

FLUSHING OF THE OIL SYSTEM

The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

Do this task if you find contamination of the oil system:

- De-energize the electrical system
- Obey all the electrical/electronic safety precautions

WARNING

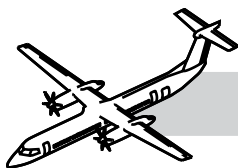
MAKE SURE THAT THE OIL IS NOT HOT WHEN YOU DO MAINTENANCE.

WARNING

DO NOT LET THE OIL TOUCH YOUR SKIN FOR A LONG TIME. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

Flush the engine oil system as follows:

- Drain the engine oil system
- Fill the engine oil system
- Start the engine
- Run the engine at 80% Nh until the oil temperature is 70 to 90 degrees C
- Make sure the oil pressure is 61 to 72 psid
- Shutdown the engine
- Drain the engine oil system
- Remove and discard the main oil filter element
- Install a new filter element
- Remove and discard the scavenge oil filter element
- Install a new filter element
- Fill the engine oil system.



Calibrated Dipstick

NOTE

Refer to Figure 79-35. Calibrated Dipstick.

The calibrated dipstick gives a visual indication of the oil level in the tank (when checked after 10 minutes of engine shutdown).

The calibrated dipstick is used during normal engine servicing. It is on the oil tank on the port side of the engine. The dipstick is attached to the oil filler cap and has markings to indicate the quantity of oil that must be added during servicing.

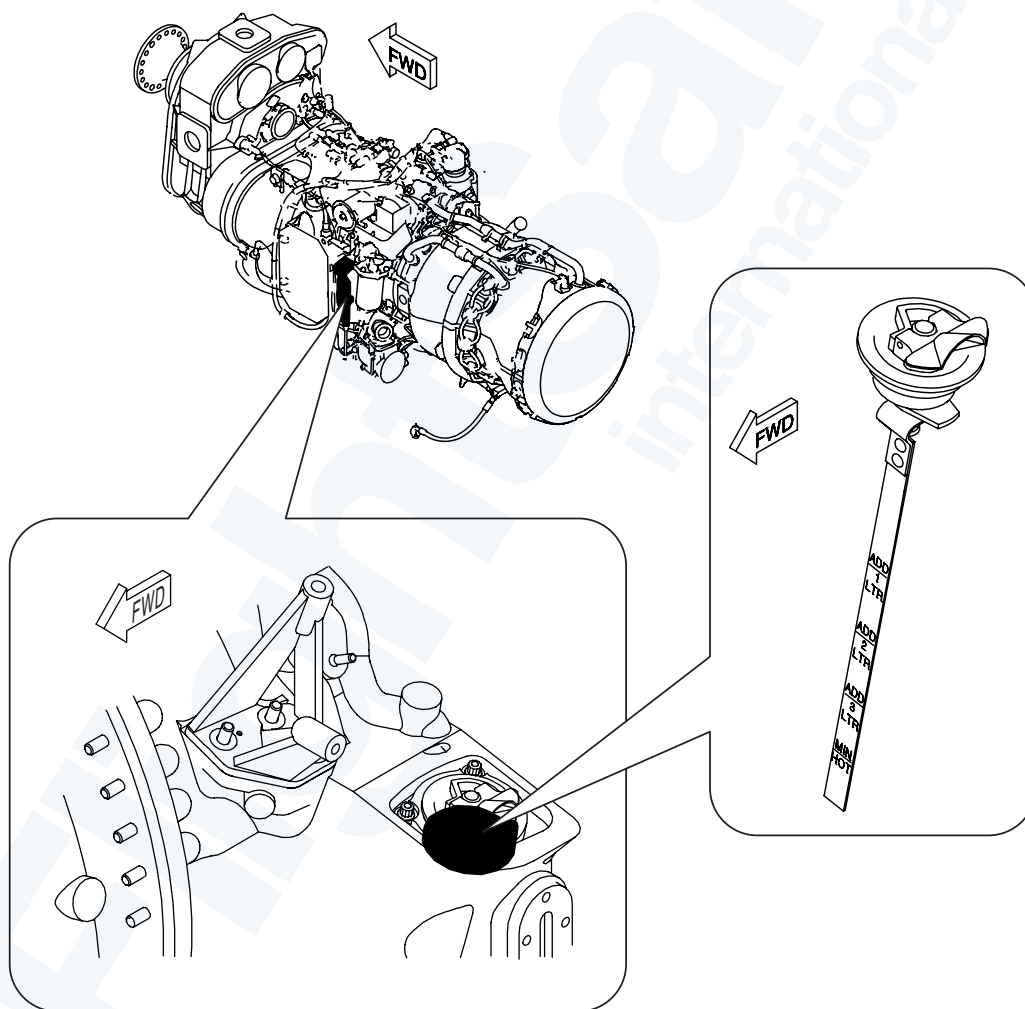
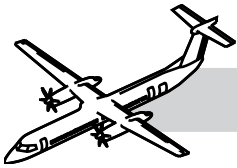


Figure 79-35. Calibrated Dipstick



Low Oil Level Indicator Glass (Bullseye)

Refer to Figure 79-36. Low Oil Level Indicator Glass (Bullseye).

The low oil level indicator glass (bullseye) gives a visual indication of oil level in the tank.

This method is used to make sure enough oil is present to start an engine that has been static for a long period.

The low oil level indicator is below the sightglass, on the left side of the engine.

The indicator is threaded into the oil tank and is sealed with a preformed packing.

The presence of oil in the bullseye indicates sufficient oil for starting.

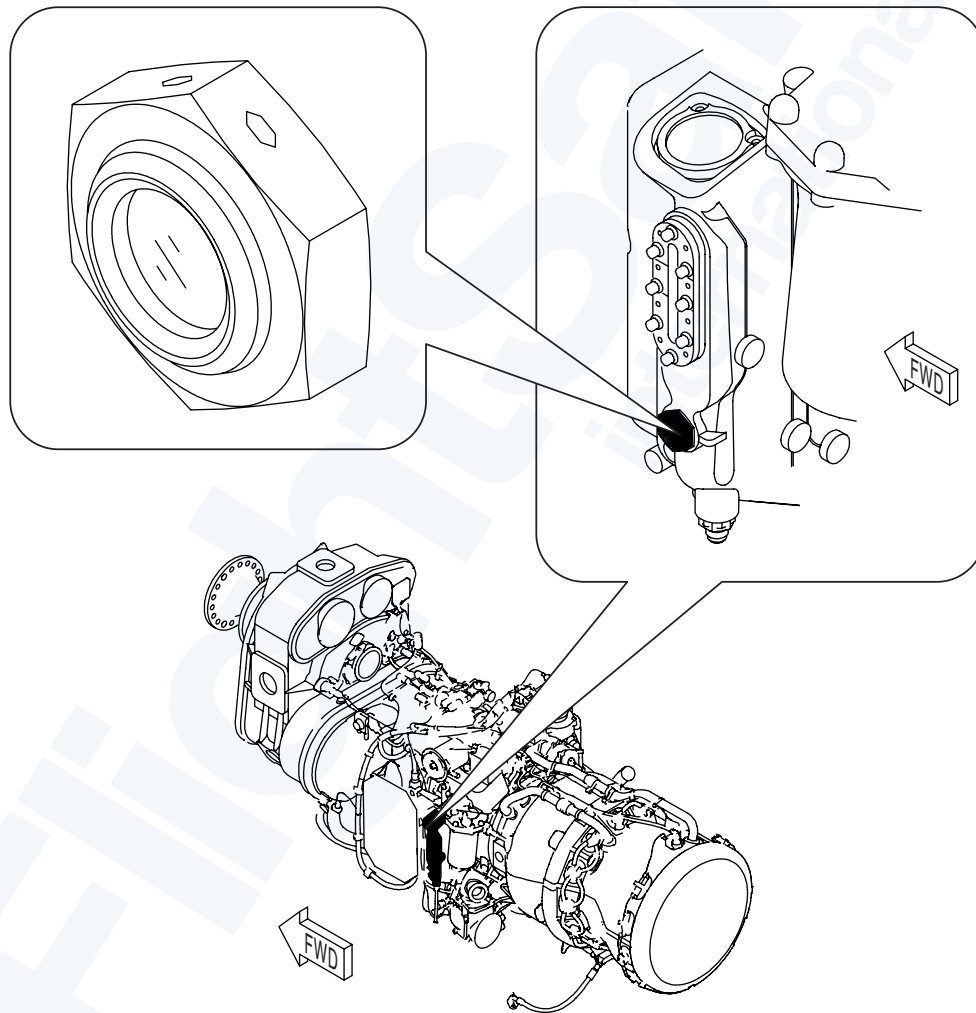
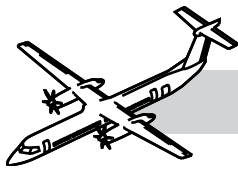


Figure 79-36. Low Oil Level Indicator Glass (Bullseye)



Remote Oil Level Indication

General

Refer to Figure 79-37. Remote Oil Indication System Block Diagram.

On aircraft with the remote oil level indication system (812CH0006), a visual indication is given in the flight compartment.

The system gives indication that there is approximately 10 flying hours or less of engine oil (based on nominal oil consumption).

The system is considered to be non-essential for operation of the aircraft.

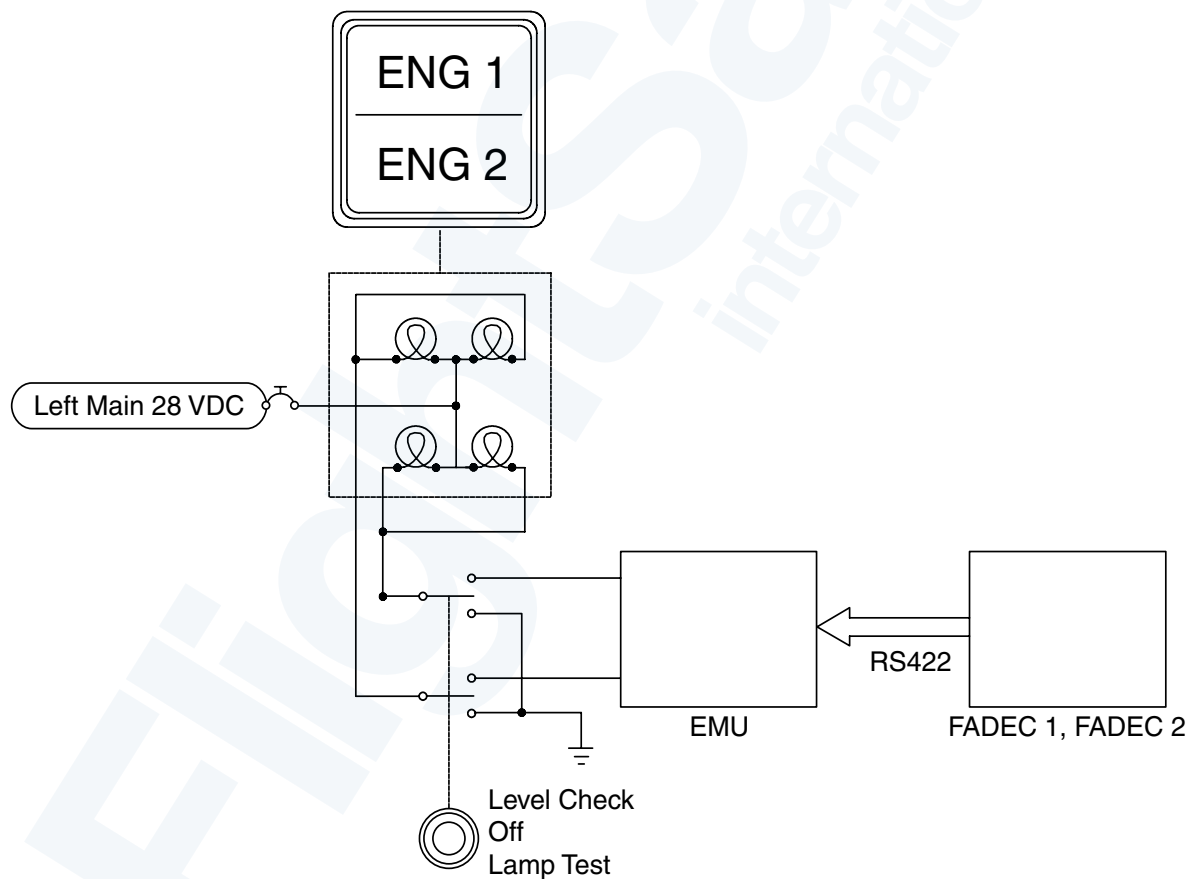
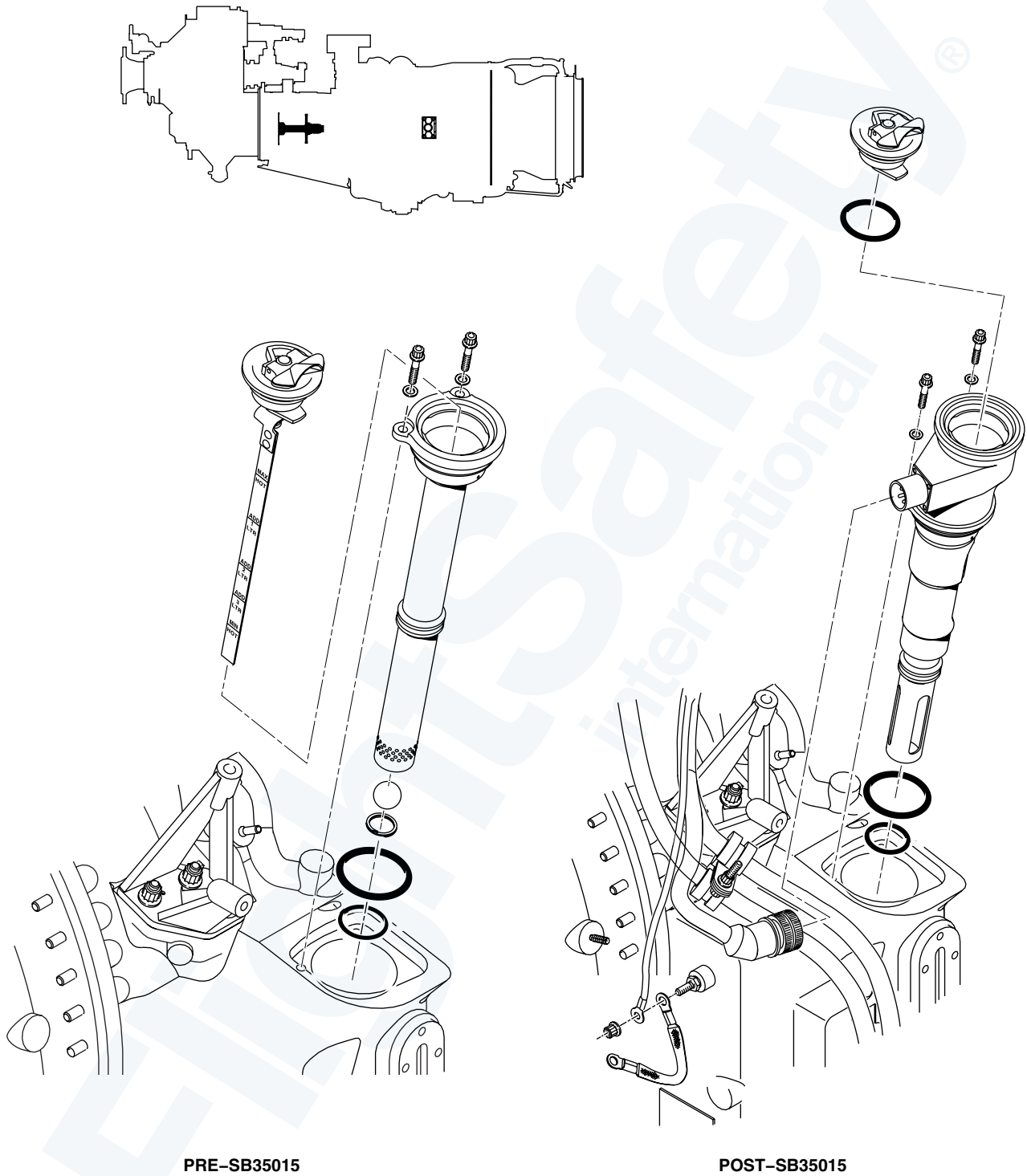
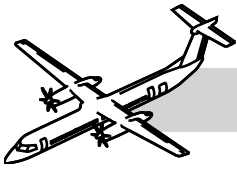
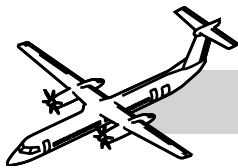


Figure 79-37. Remote Oil Indication System Block Diagram

**Figure 79-38. Oil Tank Filler Cap Installation - SB35040**



CONTROLS AND INDICATIONS

Refer to:

- Figure 79-38. Oil Tank Filler Cap Installation - SB35040.
- Table 79-1. Remote Oil Indications Chart.
- Figure 79-39. Remote Oil Level Indication - Detail.

The EMU Oil Level status is stored in the EMU so that the Oil Level Indication System can be interrogated at any time (even in flight).

This is because the system is simply giving the stored information.

However, to avoid confusion regarding the intention of the system, the control panel is placarded “For Ground Use Only”.

If more than 2.5 engine running hours have elapsed since the last successful update of the oil level status in the EMU, status will be changed to “No Valid Reading”.

Indication is given only when the system interrogation switch is selected to on.

In all cases no light indicates that maintenance action is required.

Condition	Flight Compartment Display
Oil Level OK	Steady Green Light
Oil Level Low	No Light
No Valid Reading	Flashing Green Light
EMU Switch Failed	No Light
FADEC Channel Failed	No Light

Table 79-1. Remote Oil Indications Chart

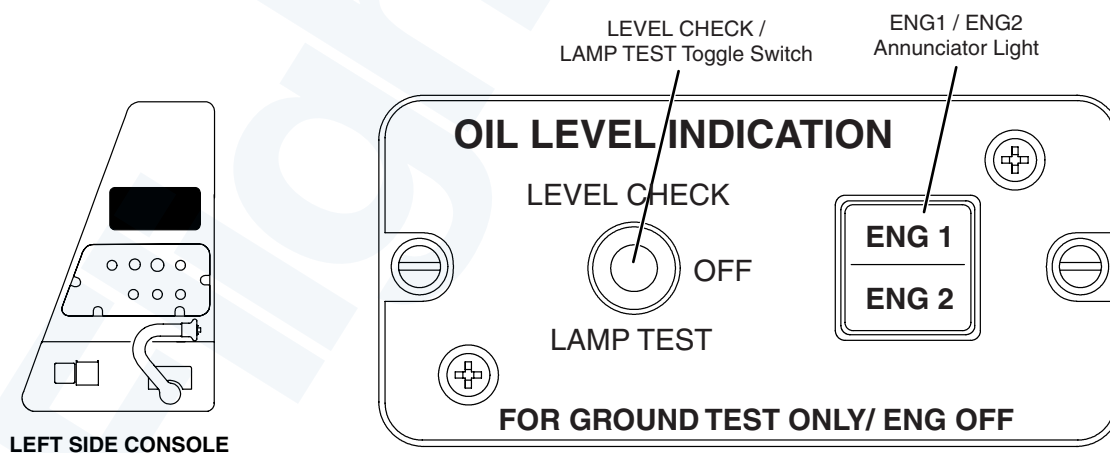
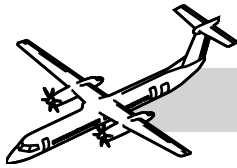


Figure 79-39. Remote Oil Level Indication - Detail



79-00-00 APPENDIX

MAINTENANCE CONSIDERATION

The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

Safety Precautions

WARNING

DO NOT LET THE OIL TOUCH YOUR SKIN FOR A LONG TIME. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION

YOU MUST FOLLOW THE INSTRUCTIONS IF IT IS NECESSARY TO CHANGE OR MIX OIL TYPES. IF YOU DO NOT DO THIS, YOU CAN CAUSE DAMAGE TO THE ENGINE.

CAUTION

STOP THE ENGINE IF THE ENGINE OIL TEMPERATURE GOES TO MORE THAN 225°F (107°C). IF YOU DO NOT DO THIS YOU CAN CAUSE DAMAGE TO THE ENGINE.

Unscheduled Inspection

Refer to the Bombardier published AMM Part 2 PSM 1-84-2.

ENGINE INSPECTION FOR FUEL IN THE OIL SYSTEM

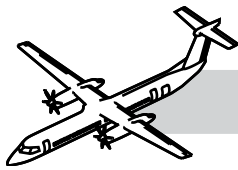
Fuel in the oil system can change the properties of the oil and cause a fire.

Do the task as follows:

- Remove the fuel heater and install a new heater
- Flush the engine oil system
- Flush the airframe oil cooler
- Remove the main and scavenge filters
- Check the filters for contamination
- Install the filters
- Check the EMU for chip detector system indication.

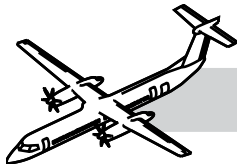
After the task is complete the EMU and the filters must be checked for contamination as follows:

- After 10 fh or 1 day of operation and
- After 15 - 35 FH and
- Again after 40 - 60 FH



79-00-00 SPECIAL TOOLS & TEST EQUIPMENT

- Solvent Bath
- Cole-Parmer P/N H- 62503-00 or Commercially Available
- Leakproof cylindrical container (Min. 8.0 in. (203.2 mm) long X 3.5 in. (88.9 mm) Dia.)
- Local Purchase Manometer
- Millipore P/Ns PD1004700 and PD1004750 Patch Receptacle (flat, circular) with cover (48 mm Min. inside Dia.)
- PWC55574 Puller
- PWC55814 Fork, Transfer Tube
- PWC57220 Screw, Jacking
- PWC42191 Puller
- PWC57186 Crimper
- PWC57392 Fork, Transfer Tube
- PWC55538 Puller
- PWC55059 Puller
- PWC55767 Base
- PWC55060 Drift
- PWC55768 Drift
- PWC55894 Knocker/Puller
- PWC55352 Drift
- PWC55957 Guide
- PWC55843 Puller
- PWC55859 Drift
- PWC64373 Base
- PWC55963 Base
- PWC57113 Drift
- PWC55896 Mandrel
- PWC56037 Drift
- PWC57349 Drift
- PWC57350 Base
- PWC55934 Base
- PWC55744 Drift
- PWC55934 Base
- PWC55956 Drift

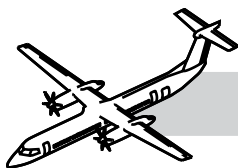


- PWC57092 Guide
- PWC64241-2 Pin, Guide
- PWC51382-460 Puller - Blowdown Valve
- PWC55829 Puller
- PWC64247 Puller
- Commercially Available Joint, Universal
- Commercially Available Megohmmeter
- Commercially Available Ohmmeter, Digital
- Commercially Available Ohmmeter
- PWC37728 Puller
- Commercially Available Megohmmeter
- Simpson (or equivalent) Ohmmeter

79-00-00 MAINTENANCE PRACTICES

Refer to the Bombardier AMM PSM 1-84-2 for details on these maintenance procedures:

- AMM 72-00-00-890-805: Oil Consumption Trend Monitoring.
- AMM 79-21-31-710-801: Operational Test of the Air-Cooled Oil Cooler Flap Door Actuator.
- AMM 79-22-31-000-801: Removal of the Oil Filler Cap and Tube Assembly.
- AMM 79-22-31-400-801: Installation of the Oil Filler Cap and Tube Assembly.
- AMM 71-00-00-780-801: Engine Oil Pressure Check and Adjustment.
- AMM 79-32-01-720-801: Functional Test of the MOT Sensor.
- AMM 79-33-01-000-801: Removal of the Oil Low Pressure Switch.
- AMM 79-33-01-400-801: Installation of the Oil Low Pressure Switch.
- AMM 79-34-00-280-801: Debris Analysis and Material Specifications.
- AMM 79-34-00-750-801: Visual Check of the Chip Detector Indicating System.
- AMM 79-34-01-700-801: Testing of the Chip Detector Magnetic Plug.
- AMM 79-34-00-720-801: Functional Test of the Chip Detection System.
- AMM 12-10-71-612-802: Filling of the Engine Oil System.
- AMM 12-10-71-617-801: Flushing of the Oil System.
- AMM 12-10-71-210-802: Engine Oil level Check.
- AMM 79-36-00-720-801: Functional Test of the Remote Oil Level Indicating System.



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